ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE
School of Basic Sciences, Institute of Physics
Laboratory of Astrophysics

On the origin of central density evolution in isolated dark matter halo simulations

Bachelor's Thesis

by

Martí Ayuso Maldonado

Supervised by

Dr. Yves Revaz

Submitted on 13 June 2026

Lausanne, Switzerland – 2026
Laboratory of Astrophysics

Abstract

This thesis investigates the origin of the persistent decline of the inner density in a collisionless dark matter halo evolved in isolation for 100 Gyr with the SWIFT code. The central question is whether the apparent cusp-to-core evolution is driven by genuine collisionless dynamics, by numerical artefacts in the gravity solver, or by post-processing choices. To address this, the thesis combines a baseline analysis of the density evolution with a set of controlled tests on the simulation volume, the initial halo structure, the gravitational softening length, and the self-gravity solver parameters. Additional diagnostics based on cumulative mass, shell decomposition, and kinematics are used to distinguish mass redistribution from numerical heating.

The results show that the inner density declines steadily throughout the full integration time, but the effect is not removed by changes in box size, softening, or solver settings. A cored initial condition still evolves toward lower central density, although with reduced amplitude, and the hybrid truncated-NFW experiments indicate that suppressing long-range forces only weakly alters the trend until the cutoff approaches the softening scale. The kinematic evolution is consistent with numerical heating, the central velocity dispersion rises while the anisotropy becomes more tangential. Taken together, these findings rule out a simple centering or analysis artefact and point instead to residual numerical heating associated with finite-N relaxation and discreteness noise as the most likely cause of the observed core formation.

Résumé

Cette thèse examine l'origine de la diminution persistante de la densité centrale dans un halo de matière noire collisionless évolué en isolation pendant 100 Gyr avec le code SWIFT. La question centrale est de savoir si l'évolution apparente du cusp vers un coeur provient d'une dynamique collisionless réelle, d'artefacts numériques dans le solveur de gravité ou de choix de post-traitement. Pour y répondre, la thèse combine l'analyse de référence de l'évolution de densité avec des tests contrôlés portant sur le volume de simulation, la structure initiale du halo, la longueur d'adoucissement gravitationnel et les paramètres du solveur de self-gravity. Des diagnostics complémentaires fondés sur la masse cumulée, la décomposition en couches sphériques et la cinématique permettent de distinguer une redistribution de masse d'un chauffage numérique.

Les résultats montrent que la densité intérieure décroît de façon continue pendant toute l'intégration, mais que cet effet n'est ni supprimé par la taille de la boîte, ni par l'adoucissement, ni par les paramètres du solveur. Une condition initiale cored évolue elle aussi vers une densité centrale plus faible, quoique avec une amplitude réduite, et les expériences hybrides truncated-NFW indiquent que la suppression des forces à longue portée ne modifie que faiblement la tendance jusqu'à ce que le rayon de coupure approche l'échelle d'adoucissement. L'évolution cinématique est compatible avec un chauffage séculaire : la dispersion de vitesse centrale augmente tandis que l'anisotropie devient plus tangente. L'ensemble de ces résultats exclut un simple artefact de centrage ou de traitement et indique plutôt un chauffage numérique résiduel, lié à la relaxation à N fini et au bruit de discrétisation, comme cause la plus probable de la formation observée du coeur.

Contents

List of Figures

1 Introduction

Dark matter makes up most of the mass in galaxies and galaxy clusters and plays a crucial role in shaping cosmic structure. Despite decades of observational and theoretical effort, its fundamental nature remains one of the outstanding questions in physics and astronomy. At cosmological scales, dark matter halos form through gravitational collapse and hierarchical assembly, producing characteristic density profiles that encode information about the underlying dark matter model, assembly history, and internal dynamics.

Numerical simulations provide a controlled setting in which these processes can be studied. Interpreting the results requires care, because subtle trends may reflect genuine physics or arise from modelling choices and post-processing. The chapters that follow develop the theoretical background, describe the computational framework, and present the diagnostics and experiments used throughout this thesis.

1.1 The cusp-core problem

In standard Cold Dark Matter (CDM) cosmology, gravitational N-body simulations predict that dark matter halos develop steep, cuspy density profiles in their central regions, following a power-law slope $\gamma \approx -1.5$ to -2 within a characteristic transition radius. Observationally, however, many dwarf galaxies exhibit much shallower central profiles with slopes closer to $\gamma \approx -0.2$ to -1, a feature referred to as a density core. This discrepancy between the predicted cusps and observed cores, known as the *cusp-core problem*, has motivated decades of investigation into whether the resolution lies in the nature of dark matter, the dynamical history of galaxies, or systematic observational uncertainties (Power et al., 2003; Zhang et al., 2019).

One promising explanation for core formation involves non-adiabatic perturbations to the potential. When supernovae, black hole feedback, or other energetic events suddenly alter the gravitational potential of a galaxy, orbits that were previously in equilibrium can be disrupted. Particles at small radii can then diffuse outward through repeated phase-space mixing, gradually flattening the inner density profile. More subtly, stochastic gravitational fluctuations generated by clumpy substructure or halo asymmetries can also transfer energy and angular momentum to central orbits over long dynamical times, producing a cumulative heating effect (Peñarrubia, 2017, 2019). Such mechanisms appear particularly efficient in ultra-faint dwarf galaxies, which have shallow potential wells and are prone to ongoing tidal interactions.

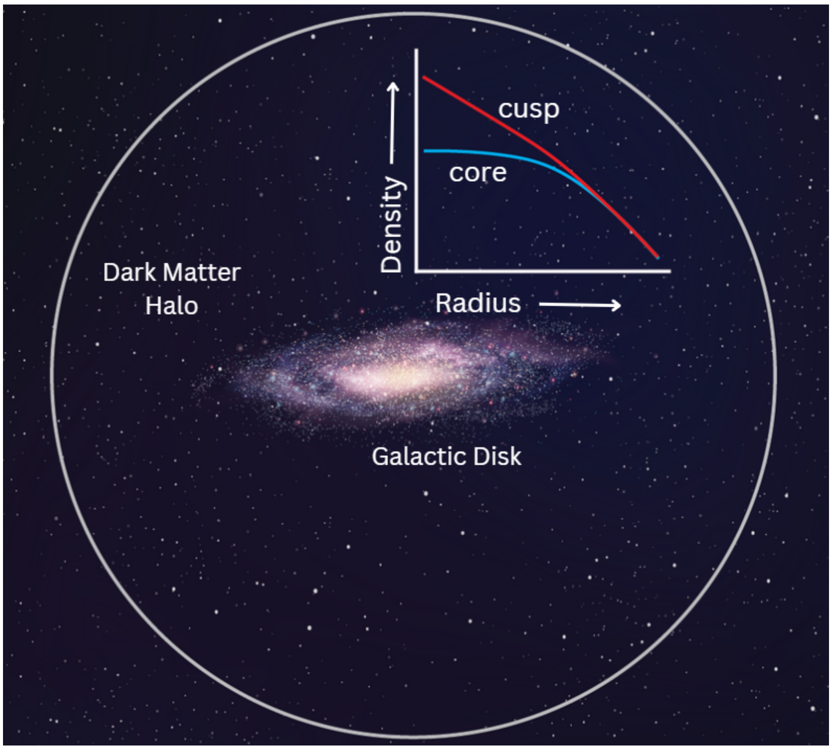


Figure 1.1: Cusp versus core density profiles. Image credit: AAS Nova (Down to the core: Dark matter deviations in ultra-faint dwarf galaxies) (AAS Nova, 2024).

1.2 Motivating question

The persistent inner-density decline observed in controlled, dark-matter-only simulations of relaxed halos presents a puzzle. In such runs, there are no baryonic processes (supernovae, AGN feedback) and no time-dependent external perturbations. Gravity is purely Newtonian, computed with modern adaptive algorithms. Yet the simulated halos still exhibit a gradual flattening of their central profiles over time. This raises a critical question: is this decline a reflection of genuine collisionless dynamics, an artifact of numerical approximations in the gravity solver, or a side effect of the post-processing analysis pipeline?

If the observed density decline is primarily a numerical artifact, then the apparent core in simulations is unreliable and informs little about real galaxy evolution. If it is a genuine dynamical feature, then understanding its mechanism opens a window into collisionless relaxation and the role of global halo coupling.

This thesis focuses on determining which of these possibilities explains the persistent central-density decline observed in a reference SWIFT simulation of a dark matter halo evolved in isolation over 100 Gyr.

1.3 Aims and objectives

The main goal of this work is to investigate the origin of the central-density decline through controlled parameter studies and tests. The specific objectives are:

1. **Characterize the baseline behavior**: Establish what the reference simulation produces in terms of density-profile evolution, and verify that the trend is robust against simple post-processing variations (centering method, snapshot noise reduction).
2. **Rule out trivial numerical explanations**: Test whether changes to numerical parameters significantly alter the observed decline. If numerical parameters prove insensitive, a more fundamental physical or structural process becomes more likely.
3. **Assess global and local influences**: Investigate whether the inner density trend reflects purely local physics or depends on the larger-scale halo environment.
4. **Evaluate the robustness of the effect**: Test whether the decline persists for different initial halo structures and different simulation volumes, to establish whether the mechanism is universal or tuning-dependent.

The cumulative outcome will be either a validated mechanism (if the trend survives all tests and correlates with a specific physical process) or a ruled-out category of explanations.

1.4 Thesis outline

The thesis is organized as follows:

Chapter 1: Theoretical background introduces the collisionless framework governing dark matter halos, defines the cusp-core phenomenology, and reviews the numerical considerations that affect halo simulations. This chapter establishes the theoretical minimum needed to interpret the results.

Chapter 2: Simulation model and implementation documents the numerical setup in detail. The SWIFT configuration, initial condition generation, halo parameter choices, and implementation details of the truncated-NFW model are described. This chapter is designed for reproducibility, a reader should be able to replicate the simulations and analysis from the information provided.

Chapter 3: Results and discussion presents the parameter studies and sensitivity analyses. The chapter progresses through five main tests: (i) the baseline evolution, (ii) box-size sensitivity, (iii) initial-condition dependence (cusp versus core), (iv) softening-length variations, and (v) self-gravity solver parameter sensitivity. Each test is framed as a hypothesis to either support or eliminate candidate explanations. Later sections address shell contributions and hybrid models.

Chapter 4: Conclusions summarizes which explanations have been ruled out and which remain plausible, and outline suggestions for future work.

This thesis therefore focuses on determining whether the effect is numerical, methodological, or physical in origin, by combining targeted diagnostics with controlled variations of the numerical setup and initial conditions.

2 Theoretical background

This chapter introduces the physical and numerical concepts required to interpret the evolution of the inner dark matter density in ultra-faint dwarf galaxy simulations. The objective is not to review all of modern structure formation, but to establish the framework needed for the analyses in subsequent chapters: collisionless halo dynamics, cusp/core phenomenology, numerical heating, gravitational softening, the gravity solver controls used in SWIFT, and the role of shell-to-shell coupling in mass redistribution.

2.1 Dark matter halos and density profiles

In the collisionless limit, dark matter is described by a phase-space distribution function $f(\mathbf{x}, \mathbf{v}, t)$ that obeys the collisionless Boltzmann equation, coupled to Poisson's equation for the gravitational potential (Binney and Tremaine, 2008). The key implication is that halo evolution is governed by a smooth mean gravitational field rather than by direct microscopic collisions between particles.

Using standard notation, the collisionless system is governed by

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f - \nabla_{\mathbf{x}} \Phi \cdot \nabla_{\mathbf{v}} f = 0, \tag{2.1}$$

with the potential determined by

$$\nabla^2 \Phi = 4\pi G \rho, \qquad \rho(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{v}, t)\, \mathrm{d}^3\mathbf{v}. \tag{2.2}$$

These equations make clear that halo structure is a phase-space problem: the density profile cannot be understood independently of the velocity distribution and orbital content.

In cosmological simulations, dynamically relaxed halos are commonly described by broken power-law density profiles, where the logarithmic slope changes from inner to outer radii. In this language, the central shape of the profile is the critical diagnostic, a steep rise toward small radii is referred to as a *cusp*, while a flattened inner profile is referred to as a *core*. Since this inner region is also the most demanding numerically, any physical interpretation must be tied to explicit convergence and resolution arguments (Power et al., 2003; Zhang et al., 2019).

A widely used fitting form for collisionless dark matter haloes is the Navarro–Frenk–White (NFW) profile,

$$\rho(r) = \frac{\rho_s}{(r/r_s)\,(1 + r/r_s)^2}, \tag{2.3}$$

where ρ_s is a characteristic density and r_s is the scale radius at which the logarithmic slope changes from approximately -1 to -3.

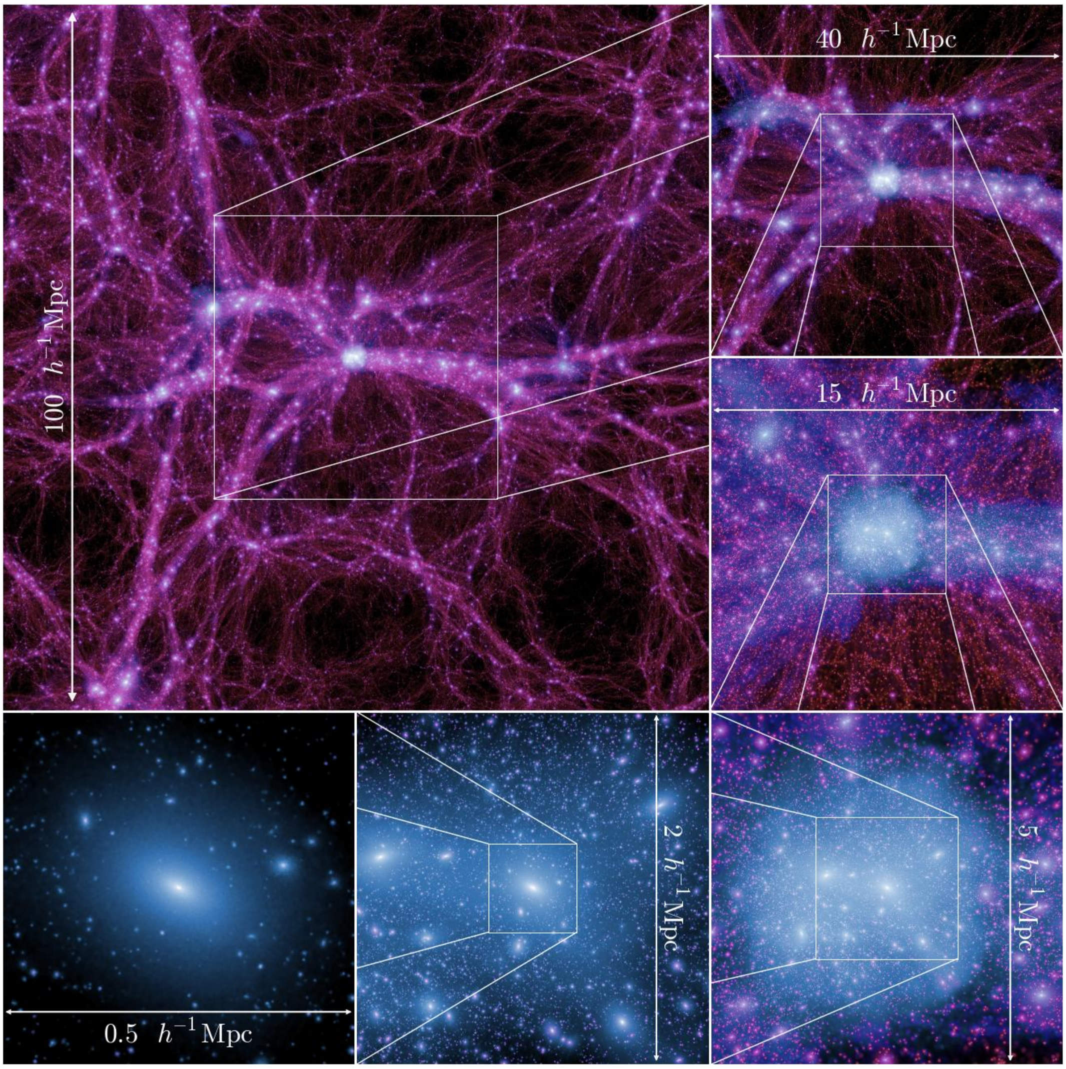


Figure 2.1: Zoom sequence (100 to 0.5 Mpc/h) into the most massive halo at $z = 0$ from the Millennium-II Simulation (Boylan-Kolchin et al., 2009).

2.1.1 Virial equilibrium, characteristic timescales, and orbital support

Dark matter haloes are not static objects, but many relaxed systems are close enough to dynamical equilibrium that a virial description is useful.

For a virialized self-gravitating system, the time averaged kinetic and potential energies approximately satisfy

$$2K + U \approx 0, \tag{2.4}$$

which provides a useful global characterization of dynamical equilibrium.

In that regime, the balance between kinetic and potential energy sets the overall structure of the halo, while the local orbital distribution determines how the inner profile evolves over time. The relevant timescale is the dynamical time,

$$t_{\rm dyn}(r) \sim \left[G\bar{\rho}(< r)\right]^{-1/2}, \tag{2.5}$$

which becomes short in the dense central regions and therefore makes the inner halo especially sensitive to both physical perturbations and numerical error.

A useful way to connect density structure to orbital support is through the spherical Jeans equation,

$$\frac{\mathrm{d}\left(\rho\sigma_r^2\right)}{\mathrm{d}r} + 2\beta(r)\frac{\rho\sigma_r^2}{r} = -\rho\frac{\mathrm{d}\Phi}{\mathrm{d}r}, \tag{2.6}$$

where σ_r is the radial velocity dispersion and $\beta(r)$ is the anisotropy parameter. This equation shows that the density profile cannot be interpreted independently of the velocity structure: two haloes with similar density profiles may differ substantially in orbital content and stability.

Radial anisotropy is particularly important because systems dominated by radial orbits can respond differently to perturbations and secular evolution than more isotropic systems. Changes in orbital anisotropy may therefore accompany or even drive changes in the inner density structure.

2.2 Cusp-core behaviour in ultra-faint dwarf galaxies

Ultra-faint dwarf galaxies are particularly valuable for small-scale dark matter studies because they are strongly dark matter dominated and have shallow gravitational potentials. Their internal structure is therefore sensitive both to the underlying dark matter model and to dynamical heating processes that can redistribute mass over time (Salamin, 2026).

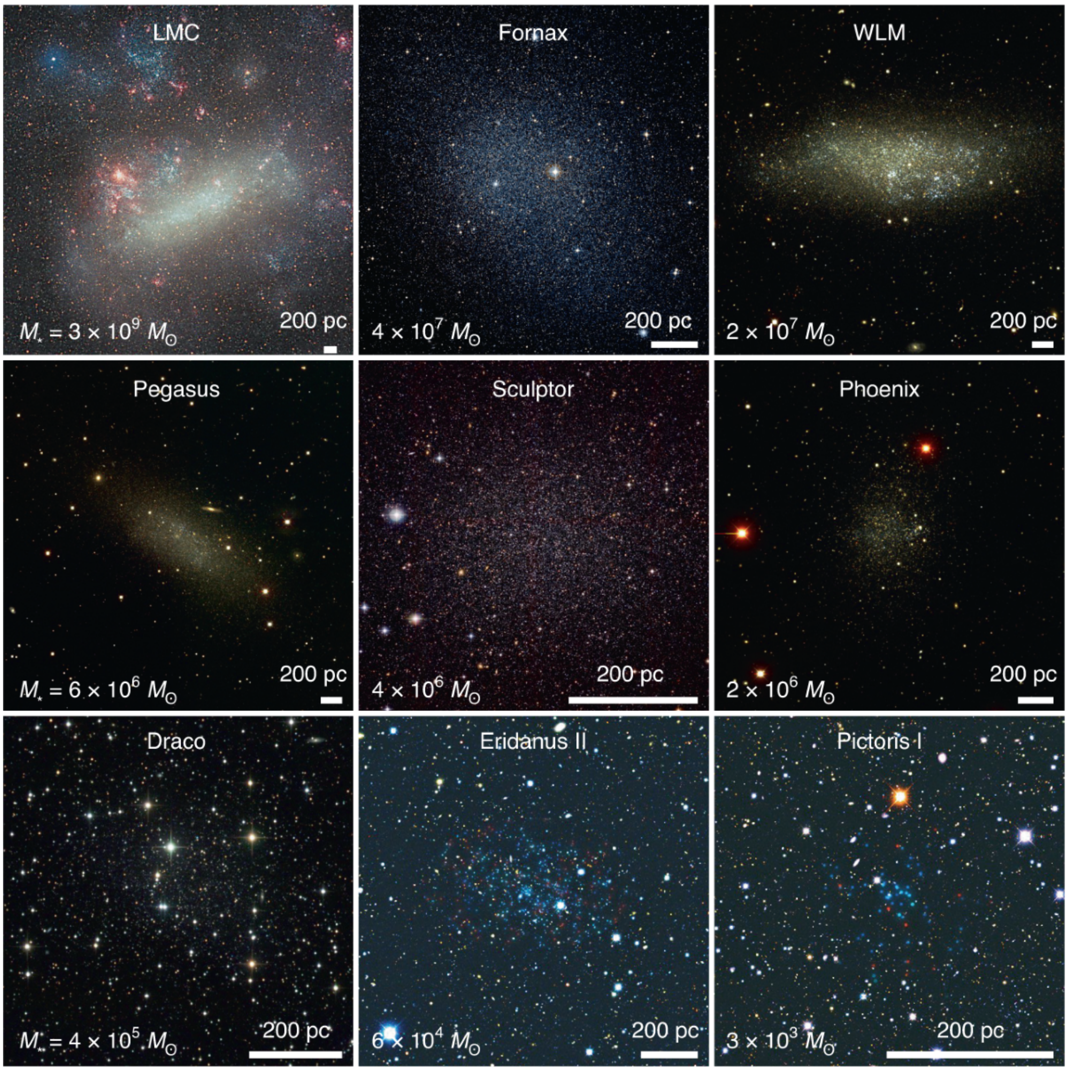


Figure 2.2: Collection of observed dwarf galaxies (Crnojević and Mutlu-Pakdil, 2021).

In a cold dark matter scenario, cuspy inner profiles are the usual expectation for collisionless halos. In warmer dark matter scenarios, free-streaming suppresses small scale power and can alter assembly histories, potentially affecting central structure indirectly through different accretion and merger pathways (Salamin, 2026). Independently of the particle model, repeated perturbations and non-adiabatic potential fluctuations can transfer energy to central orbits and drive profile flattening.

The conceptual point is that core formation does not originate from one mechanism. Similar-looking inner profiles can emerge from different routes: altered initial fluctuation spectra, stochastic heating by external perturbers, or numerical relaxation. As a result, interpretation requires combining profile evolution with complementary diagnostics.

This creates a central interpretive challenge for the present work. An apparent cusp-to-core transition can arise from physical mechanisms, from numerical artifacts, or from both acting simultaneously.

2.3 Numerical heating and relaxation effects in N-body systems

An N-body simulation represents a continuous collisionless fluid with a finite number of massive macro particles. This discretization introduces spurious collisionality, close encounters can produce artificial two-body relaxation, velocity diffusion, and long-term heating, especially in poorly resolved central regions (Binney and Tremaine, 2008; Power et al., 2003).

To first order, the local relaxation timescale scales approximately as

$$t_{\mathrm{relax}} \sim \frac{N}{8 \ln N}\, t_{\mathrm{cross}}, \tag{2.7}$$

where N is the number of effective particles supporting the region of interest and t_{cross} is a characteristic crossing time (Binney and Tremaine, 2008). This scaling emphasizes why central halo regions are vulnerable: they contain fewer particles and shorter dynamical times, making discreteness effects comparatively stronger. If the numerical relaxation time at a given radius is not much longer than the simulated evolution time, then the resulting structure may be numerically biased.

Power et al. (2003) translated these ideas into practical convergence guidance for halo simulations by linking reliability in the inner profile to a combination of mass resolution, timestep sampling, and force softening. Their framework remains a benchmark because it moves beyond single-parameter tuning and treats convergence as a coupled numerical problem.

2.4 Gravitational softening: fixed and adaptive formulations

Gravitational softening modifies the short range Newtonian force to prevent unphysical singular accelerations from particle-particle encounters. With a fixed softening length ϵ, one obtains a trade-off, large ϵ suppresses noise and relaxation but erases genuine small scale structure; small ϵ improves nominal resolution but can increase heating (Power et al., 2003; Zhang et al., 2019).

In many implementations, this can be written schematically with a softened pair force,

$$\mathbf{F}_{12} = -\frac{G m_1 m_2}{\left(r_{12}^2 + \epsilon^2\right)^{3/2}}\, \mathbf{r}_{12}, \tag{2.8}$$

which regularizes the r^{-2} singularity at very short separation (Zhang et al., 2019). The interpretation of ϵ is therefore both a numerical stabilizer and a scale that limits physically meaningful force gradients.

Power et al. established influential convergence criteria linking softening, timestep, and particle number in the inner halo (Power et al., 2003). Later work showed that these prescriptions may be conservative for some setups, while also emphasizing that excessively small softening can bias the innermost density high (Zhang et al., 2019). Taken together, these studies show that softening is not a single best number independent of context.

Adaptive softening addresses this by varying effective force resolution with local conditions. However, straightforward implementations do not preserve conservation laws. The Lagrangian formalism of Price and Monaghan (2007) demonstrated how to construct adaptive softening with the additional correction terms required for consistent energy and momentum conservation. Modern conservative extensions generalize this framework to multi-species collisionless simulations (Hopkins et al., 2023).

From a methodological perspective, adaptive schemes are especially relevant for halos with strong radial density contrasts. A fixed ϵ that is adequate in the outskirts may be too coarse in the center, while an ϵ tuned to the center may be too noisy at large radii. Conservative adaptive formulations provide a way to mitigate this tension while preserving the invariants required for long integrations.

In the context of this thesis, softening is a key control variable which will be studied in detail.

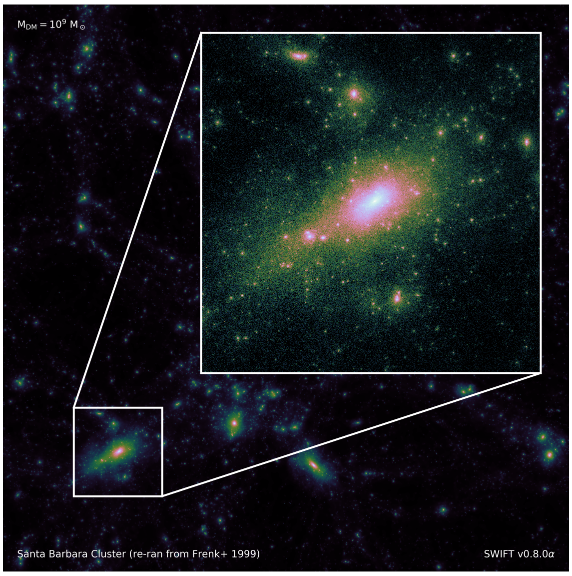
2.5 Gravity solver controls in SWIFT

Beyond softening, force computation in modern cosmological codes depends on hierarchical gravity solvers (tree and fast multipole expansions) and their acceptance criteria. These methods accelerate long range gravity by replacing exact pairwise summation with controlled approximations, whose errors depend on opening criteria and expansion tolerances.

This introduces a second numerical layer on top of softening. Even with identical initial conditions and identical softening, changes in opening criteria or multipole tolerances can alter force errors in a way that accumulates over many dynamical times.

In practical terms, parameters such as the multipole acceptance criterion (MAC), geometric opening thresholds (e.g. θ_{cr}), expansion tolerances (e.g. FMM error controls), and timestep factors set the balance between computational cost and force accuracy. Because central density evolution is sensitive to small cumulative force biases, these controls must be included in any sensitivity analysis of cusp/core evolution.

Agora galaxy (GEAR model). A simulation highlighting gas, star formation and morphological features produced with the GEAR physics module. Image: Yves Revaz, SWIFT gallery.



Santa-Barbara Cluster. A large-scale cluster formation snapshot from the Santa Barbara comparison suite. Image: Josh Borrow, SWIFT gallery.

Figure 2.3: Simulation examples from the SWIFT gallery (SWIFT Collaboration, 2024).

2.6 Shell structure and mass redistribution

central potential is influenced by mass and orbital structure over a broad radial range. Particle populations at larger radii can contribute to central acceleration fluctuations, angular momentum exchange, and gradual redistribution of enclosed mass. Recent theoretical work on stochastic tidal fields generated by clumpy substructure provides a useful interpretation for this non-local coupling. In these treatments, gravitational fluctuations induce diffusion in orbital integrals and can transfer energy and angular momentum through repeated perturbations rather than through single events (Peñarrubia, 2019, 2017).

An important implication is that central density decline may be driven by the full halo environment, not only by the innermost force law. If true, changing local softening alone may leave a substantial part of the trend unchanged. This non-local perspective is consistent with previous findings in the project context, where persistent central density decline can remain visible across multiple numerical variations (Salamin, 2026).

3 Simulation model and implementation

This chapter describes the numerical setup used to investigate the persistent central density decline identified in the previous chapters. The focus is on reproducibility: each subsection documents the simulation code configuration, initial condition choices, and implementation details used to test numerical and physical explanations.

The workflow combines (i) collisionless N-body runs in SWIFT, (ii) profile diagnostics extracted from snapshots, and (iii) a hybrid implementation designed to quantify non-local contributions to the dynamics of the innermost region.

3.1 SWIFT simulation setup

All simulations in this thesis are dark matter only and non-periodic, evolved with SWIFT in self-gravity mode (and external gravity when required by specific tests). In the working setup, runs are launched through

```
swift --self-gravity [--external-gravity] params.yml
```

with outputs consisting of snapshots, global statistics, and timestep logs.

The internal unit system is chosen so that

$$1 \text{ length unit} = 1 \text{ kpc}, \qquad 1 \text{ velocity unit} = 1 \text{ km s}^{-1}, \qquad 1 \text{ mass unit} \approx 10^{10}\, M_\odot, \tag{3.1}$$

which allows straightforward interpretation of profile radii (kpc), velocity dispersions (km/s), and enclosed masses in solar units after post-processing conversion.

3.1.1 Baseline gravity and time integration choices

The parameter-study baseline adopts the self-gravity configuration

$$\eta = 0.002, \qquad \mathrm{MAC} = \text{geometric}, \qquad \theta_{\mathrm{cr}} = 0.7, \qquad \epsilon_{\mathrm{fmm}} = 10^{-3}, \tag{3.2}$$

with minimum and maximum timesteps typically constrained in the interval

$$\Delta t_{\min} = 10^{-6}, \qquad \Delta t_{\max} = 10^{-2} \tag{3.3}$$

To probe solver sensitivity, each of the following was varied while keeping the others fixed: timestep factor η, MAC choice (geometric/adaptive), geometric opening threshold θ_{cr}, and ϵ_{fmm}.

3.1.2 Output cadence and bookkeeping

The production analysis relies on regular snapshot dumps that allow profile reconstruction at multiple times. In parallel, SWIFT writes global conservation and bulk quantities in `statistics.txt`, and timestep activity in `timesteps.txt`.

3.2 Halo initial conditions and parameter choices

The halo models are generated as collisionless particle realizations with controlled inner slopes to enable cusp/core comparisons under otherwise matched numerical conditions.

Initial conditions are produced with the pNbody code:

```
ic_gen_2_slopes --alpha 1 --beta 3 --Rmin 1e-3 --Rmax 50 --r_cutoff 50 \
    --rho0 0.0027658297046461644 --rs 0.828177997155062 \
    --ptype 1 --mass 61664.060524 -t swift -o halo.hdf5
```

which corresponds to a two-slope halo with tunable inner structure and an outer truncation radius.

The most relevant controlled parameters are:

- halo type: cusp and core initial conditions,
- box-size variants for finite-volume sensitivity checks,
- gravitational softening variants,
- self-gravity solver parameters (η, MAC, θ_{cr}, ϵ_{fmm}).

The practical philosophy is to keep one baseline run and modify one parameter family at a time, so each subsection in the results can be interpreted as a direct hypothesis test rather than a multi-parameter confounded comparison.

3.2.1 Simulation campaign summary

Rather than repeating the qualitative run descriptions from Chapter 5, this subsection defines the campaign as a technical design matrix. The baseline control vector is

$$\mathbf{p}_0 = \{\eta = 2\times10^{-3},\ \mathrm{MAC} = \text{geometric},\ \theta_{\rm cr} = 0.7,\ \epsilon_{\rm fmm} = 10^{-3},\ \epsilon_{\rm soft} = 0.01\,\mathrm{kpc}\}, \tag{3.4}$$

and each run family perturbs exactly one component (or one model block) relative to $\mathbf{p}_0$.

Table 3.1: Technical campaign matrix written as perturbations of the baseline control vector $\mathbf{p}_0$. $N_{\rm lev}$ is the number of tested levels in each sweep; "range factor" denotes the span between minimum and maximum tested numeric values.

Control family	**IC**	**Control-vector form**	$N_{\rm lev}$	**Range factor**
Baseline reference	Cuspy	$\mathbf{p} = \mathbf{p}_0$, baseline box	1	–
Box-size	Cuspy	$\mathbf{p} = \mathbf{p}_0$, $L_{\rm box} \in \{L_{\rm s}, L_{\rm m}, L_{\rm l}\}$	3	n/a
Initial structure	Cuspy, core	$\mathbf{p} = \mathbf{p}_0$, IC$\in$ {cuspy, core}	2	n/a
Softening	Cuspy	$\mathbf{p} = \mathbf{p}_0$, $\epsilon \in \{1, 5, 10, 50, 100\}$ pc	5	100
Solver: timestep	Cuspy	$\mathbf{p} = \mathbf{p}_0$ with $\eta \in \{2\times10^{-4}, 2\times10^{-3}, 2\times10^{-2}\}$	3	10^2
Solver: MAC	Cuspy	$\mathbf{p} = \mathbf{p}_0$ with MAC$\in$ {geometric, adaptive}	2	n/a
Solver: opening angle	Cuspy	$\mathbf{p} = \mathbf{p}_0$ with $\theta_{\rm cr} \in \{0.5, 0.6, 0.7\}$ (geometric MAC)	3	1.4
Solver: FMM tolerance	Cuspy	$\mathbf{p} = \mathbf{p}_0$ with $\epsilon_{\rm fmm} \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$	4	10^3
Hybrid truncated-NFW	Cuspy	$\mathbf{p} = \mathbf{p}_0$, model→hybrid, $R_{\rm cut} \in \{50, 25, 10, 5, 3, 2, 1\}$ kpc + vanilla	8	50

This representation makes the campaign auditable at a glance: all families are written as controlled perturbations of $\mathbf{p}_0$, with sweep spans of 10^2 in η, 10^3 in $\epsilon_{\rm fmm}$, 10^2 in softening around the baseline $\epsilon_{\rm soft} = 0.01\,\mathrm{kpc}$ (1–100 pc), and 50 in $R_{\rm cut}$. A persistent trend across these families is therefore difficult to attribute to a narrow parameter-tuning artefact.

3.3 Numerical implementation of the hybrid model

A dedicated hybrid gravity model was implemented and tested on a custom SWIFT branch (`truncated-NFW`). The model enforces a hard radial split at a cutoff radius $R_{\rm cut}$:

- inner region ($r \le R_{\text{cut}}$): standard self-gravity, no external NFW potential,
- outer region ($r > R_{\text{cut}}$): external NFW potential applied, with optional masking of dark matter self-gravity sourcing outside the cutoff.

3.3.1 Implementation objectives and design choices

The truncated-NFW backend is designed to isolate the dynamical role of the outer halo while keeping the inner region fully self consistent. The implementation follows three guiding principles:

- **Clear separation of responsibilities**: inner particles ($r \le R_{\text{cut}}$) evolve under the usual SWIFT self-gravity solver and provide the only resolved self-gravity sources inside the cutoff. Outer particles ($r > R_{\text{cut}}$) evolve in the analytic truncated-NFW field, with resolved gravity contributions removed by construction.
- **Minimal invasive changes**: the feature is implemented as an optional, compile-time selectable external potential backend and does not alter the default gravity code paths when disabled.
- **Deterministic boundary convention**: the hard boundary is defined as inner for $r \le R_{\text{cut}}$ and outer for $r > R_{\text{cut}}$, avoiding ambiguous behaviour at the cutoff.

3.3.2 Implementation overview

The truncated-NFW backend is easiest to interpret as a two stage pipeline. First, the parameter parser establishes the cutoff radius, halo center, and masking state. Second, the gravity kernels combine source side masking with target side write-back suppression, so that inner particles retain the usual resolved self-gravity while outer particles are advanced through the analytic truncated-NFW field.

Figure 3.1 summarizes this flow at a glance.

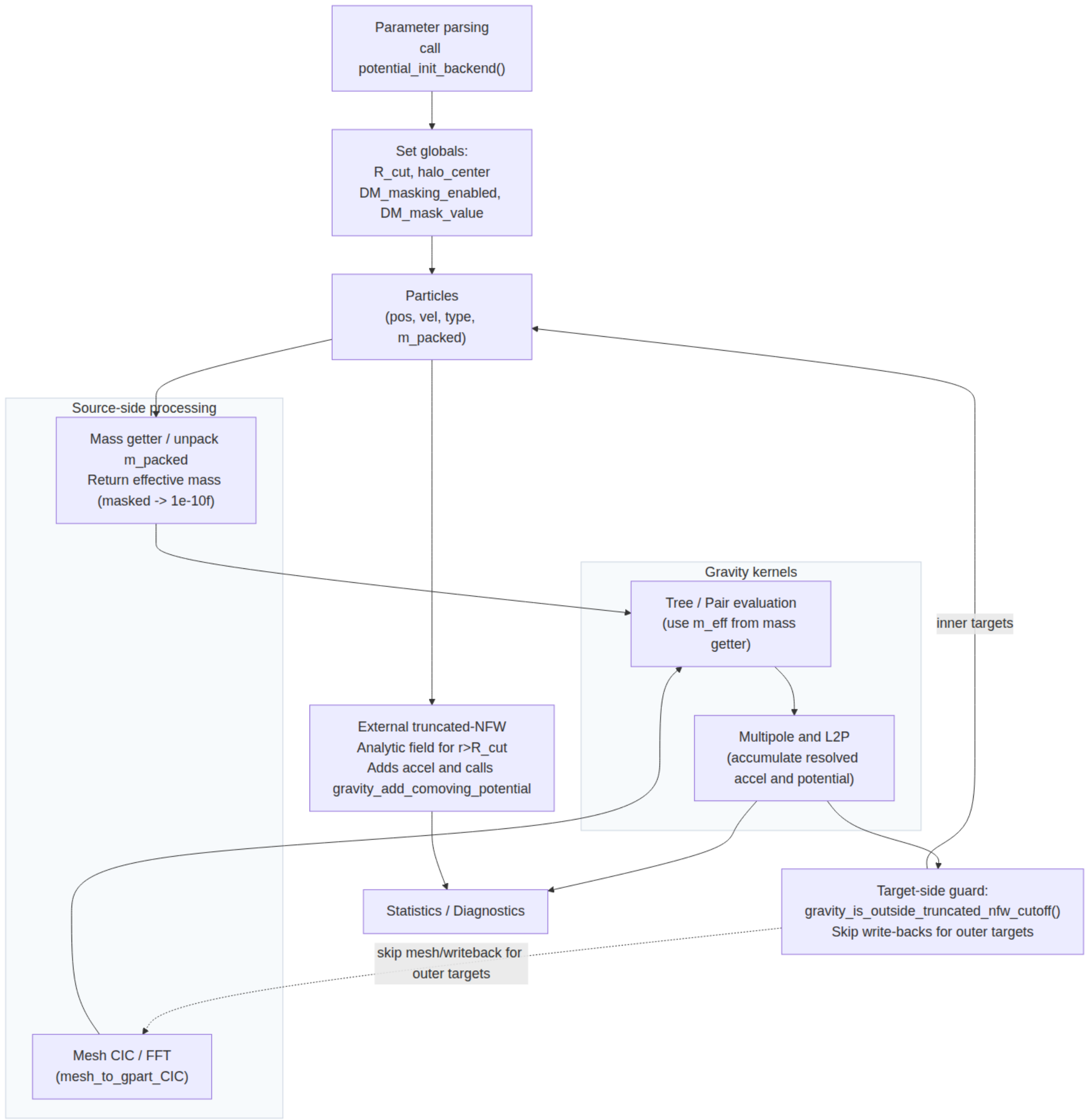


Figure 3.1: High-level control-flow schematic of the truncated-NFW implementation. The figure highlights the split between parameter initialization, source-side masking, gravity-kernel evaluation, target-side write-back suppression, and the external analytic field used for outer particles.

3.3.3 Build-time wiring and backend selection

The backend is registered as an external potential option and selected at configure time. The recommended configure invocation is:

```
./configure --with-ext-potential=truncated-nfw
```

At build-time the configure script defines the macro identifying the truncated-NFW backend and compiles the corresponding header implementation. This approach keeps the extra code path isolated behind a single configure flag and reduces the risk of accidental behavior changes in other backends.

Files involved at build time include the project's `configure.ac` (backend option and macro definition), `src/potential.h` (backend include branch), and `src/potential/truncated_nfw/potential.h` (backend implementation). Gravity-side hooks live in `src/gravity/*`, `src/multipole.h`, and mesh-gravity files, but remain dormant unless the truncated-NFW backend is selected.

3.3.4 Runtime parameters and YAML configuration

Runtime control is provided through the existing `NFWPotential` block in the parameter YAML. The truncated-NFW mode extends the canonical NFW keys with a small set of required and optional keys:

- `cutoff_radius:` required floating-point value; the radial cutoff R_{cut} in kpc.
- `dm_mask_outside_cutoff:` optional boolean (default 1). If true, DM outside the cutoff is excluded as a source of self-gravity.
- `position:` center of the external potential (array); default behaviour follows the usual `NFWPotential:useabspos` semantics.
- `useabspos:` boolean as in the standard NFWPotential, controls interpretation of `position`.

The parameter parser in the backend validates that `cutoff_radius` is present when truncated-NFW is selected and prints the effective setup at startup (center, halo parameters, cutoff, masking flag). The same center definition is propagated to the masking logic through dedicated globals (mask enabled flag, mask center, and R^2_{cut}), so potential and masking use exactly the same geometric reference.

At a mathematical level, the backend defines two indicator functions that govern the hybrid split. Let

$$\chi_{\mathrm{out}}(\mathbf{x}) = H\big(|\mathbf{x}-\mathbf{x}_c|^2 - R^2_{\mathrm{cut}}\big), \qquad \chi_{\mathrm{in}}(\mathbf{x}) = 1 - \chi_{\mathrm{out}}(\mathbf{x}), \tag{3.5}$$

where H is the Heaviside step function and $\mathbf{x}_c$ is the halo center. The implementation then acts on sources and targets through the complementary weights

$$w_{\mathrm{src}}(\mathbf{x}) = \chi_{\mathrm{in}}(\mathbf{x}) + \varepsilon_m\,\chi_{\mathrm{out}}(\mathbf{x}), \qquad w_{\mathrm{tgt}}(\mathbf{x}) = \chi_{\mathrm{in}}(\mathbf{x}), \tag{3.6}$$

with $\varepsilon_m = 10^{-25}$ for masked dark-matter sources.

The operational logic can therefore be summarized compactly as

$$\mathbf{a}(\mathbf{x}_t) = w_{\rm tgt}(\mathbf{x}_t)\,\mathbf{a}_{\rm res}(\mathbf{x}_t) + \chi_{\rm out}(\mathbf{x}_t)\,\mathbf{a}_{\rm ext}(\mathbf{x}_t), \tag{3.7}$$

where $\mathbf{a}_{\rm res}$ is the resolved contribution and $\mathbf{a}_{\rm ext}$ is the analytic truncated-NFW acceleration. In parallel, the source term entering the gravity solver is replaced by

$$m_{\rm eff}(\mathbf{x}_s) = m_s\,w_{\rm src}(\mathbf{x}_s), \tag{3.8}$$

so the tree and multipole moments are built from the masked mass field rather than from the raw particle masses. Thus, for a dark-matter particle p one may write

$$m_{{\rm eff},p} = m_p\left[\chi_{\rm in}(\mathbf{x}_p) + \varepsilon_m\,\chi_{\rm out}(\mathbf{x}_p)\right], \tag{3.9}$$

while all non dark matter species retain their original masses. The target-side rule behaves similarly: for $\chi_{\rm out}(\mathbf{x}_t) = 1$, resolved contributions from the tree, pair and mesh solvers are suppressed at write-back time, and only the analytic field contributes to the particle acceleration and external potential bookkeeping.

3.3.5 Two-layer masking logic

The gravity split is enforced through two complementary mechanisms: source-side masking (particles outside the cutoff contribute negligibly to resolved sources) and target-side suppression (outer particles do not receive resolved write-backs).

Source-side masking. To avoid expensive per-step global rewrites, the code now performs a one-time startup masking sweep that sets the stored particle mass for eligible dark matter particles to a tiny non-zero value. Concretely, a `mass_initial` field was added to the `gpart` layout (see `src/gravity/*/gravity_part.h`), and the helper `gravity_sync_truncated_nfw_masses(struct gpart *gparts, size_t)` is invoked at initialization and after restart (see `src/engine.c`). This helper applies $m_{\rm stored} = m_{\rm initial} \times \varepsilon_m$ for dark matter particles with $r > R_{\rm cut}$, where $\varepsilon_m \approx 10^{-25}$. After this one-time pass, the gravity code reads the stored `gp->mass` (and packed/foreign mass fields) directly during multipole builds, pair interactions and mesh depositions.

Note: attempts to restore or resync masses every step (or per drift) were investigated but removed because they produced unacceptable costs in typical runs.

Target-side suppression. The implementation still prevents outer particles from receiving resolved write-backs: the geometric predicate `gravity_is_outside_truncated_nfw_cutoff(...)` is used to gate the write-back paths so that, for targets with $r > R_{\rm cut}$, tree/pair/mesh contributions are not added to the particle acceleration or potential. This predicate is used in the following places:

- `src/gravity_cache.h`: in `gravity_cache_write_back(...)` a guard skips adding cached pair/tree acceleration and potential to outer targets;

- `src/multipole.h`: in `gravity_L2P(...)` an early return prevents multipole target updates for outer particles;
- `src/mesh_gravity.c`: in `mesh_to_gpart_CIC(...)` an early return suppresses local PM interpolation for outer particles;
- `src/mesh_gravity_mpi.c`: in `mesh_patch_to_gparts_CIC(...)` an early return suppresses MPI PM interpolation for outer particles.

Operationally, outer particles therefore evolve only under the analytic external potential terms, while inner particles continue to use the fully resolved self-gravity solver.

The behaviour above can be expressed in a more detailed pseudocode that mirrors the actual runtime flow and clarifies the interactions between local data, packed/foreign frames, and the external backend.

1. **Startup / restart initialization (single sweep)**

```
# parameters read from YAML / external backend
R_cut = config.cutoff_radius
centre = config.position
epsilon_m = 1e-25   # masked mass fraction

# one-time pass over local particle array (invoked from src/engine.c)
for i in 0..nr_gparts-1:
    gp = gparts[i]
    if gp.type == DM and DM_masking_enabled and |gp.x - centre|^2 > R_cut^2:
        gp.mass_initial = gp.mass          # store original mass for bookkeeping
        gp.mass = epsilon_m * gp.mass_initial  # overwrite stored mass (cheap at runtime)
    else:
        gp.mass_initial = gp.mass  # ensure mass_initial is set consistently
end
```

2. **Per-step gravity workflow (reads stored mass only)**

```
# Tree / multipole build (local + imported foreign particles)
for each node or leaf in the tree:
    use constituent particle masses (stored) to form multipole moments

# Mesh (PM) deposition: use stored mass for depositor particles
for each mesh cell patch:
    for each source particle s in patch:
        deposit s.mass (stored) to grid

# Pairwise / short-range loops use stored source masses too
for each active pair (i,j):
    use mass_i (stored) and mass_j (stored) as source factors in kernel
```

```

# Cache & write-back (target-side suppression)
for each cached target t when writing back cached contributions:
    if gravity_is_outside_truncated_nfw_cutoff(t.x):
        # gate the write-back: do not add resolved contributions for outer targets
        continue  # a_res and Phi_res remain zero for this target
    else:
        add cached pair/tree contributions to a_res(t), Phi_res(t)
```

The short-range kernels and multipole builders see the masked masses for outer sources. Targets that lie outside the cutoff are excluded from write-back in the cache and from multipole-to-particle updates (see `src/multipole.h` early return), so they do not accumulate resolved accelerations or potentials.

3. **External potential injection for outer targets**

```
for each particle t being advanced:
    if |t.x - centre|^2 > R_cut^2:
        a_res = 0, Phi_res = 0  # resolved contributions suppressed
        a_ext, Phi_ext = truncated_NFW_acc_and_pot(t.x - centre)
        a_total = a_ext + a_other_external_backends
    else:
        a_total = a_res + a_other_external_backends
```

The external backend computes analytic acceleration and potential for outer particles; an additive potential constant may be applied to keep the potential continuous at R_{cut} (this only affects reported energies, not the force).

File cross-references: the runtime pieces described above live in the following locations in the repository:

- startup masking: `src/engine.c` calls `gravity_sync_truncated_nfw_masses(...)`
- mass storage: `src/gravity/*/gravity_part.h` (`mass_initial` field)
- effective-mass helpers (diagnostic): `src/gravity/*/gravity.h` (`gravity_get_effective_mass_from_position`)
- cache write-back gating: `src/gravity_cache.h`
- multipole L2P gating: `src/multipole.h`
- mesh deposition gating: `src/mesh_gravity.c`, `src/mesh_gravity_mpi.c`
- IO bookkeeping (mass_initial set during IO/space sync): `src/common_io.c`, `src/space.c`

This pseudocode clarifies why the implementation produces the observed split in particle behaviour. Sources outside R_{cut} contribute negligibly to resolved forces (their stored mass is tiny), and outer targets are advanced only by the analytic field. Whether an individual particle remains bound after crossing depends on its instantaneous energy relative to the analytic potential at the crossing moment and is therefore a physical outcome of the hybrid model rather than an implementation artifact.

3.3.6 External potential behaviour and timestep coupling

The external potential API is responsible for returning three quantities for each particle position: acceleration $\mathbf{a}_{\text{ext}}(\mathbf{x})$, potential energy contribution $\Phi_{\text{ext}}(\mathbf{x})$, and optional timestep constraint multipliers. In truncated-NFW mode these are implemented as

- for $r \leq R_{\text{cut}}$: $\mathbf{a}_{\text{ext}} = 0$, $\Phi_{\text{ext}} = 0$ (no external field acting on inner particles),
- for $r > R_{\text{cut}}$: standard truncated NFW analytic expressions for $\mathbf{a}_{\text{ext}}$ and Φ_{ext} are returned.

The analytic potential/acceleration are injected through the standard external potential hooks (including `gravity_add_comoving_potential(...)`), while resolved-gravity suppression is handled in gravity and mesh write-back paths.

This separation clarifies potential bookkeeping. Outer particles no longer accumulate resolved self-potential from tree/pair/mesh, but they still receive the analytic external contribution via the external backend and enter $E_{\text{pot,ext}}$ diagnostics through `external_gravity_get_potential_energy(...)`.

For diagnostics, the implementation effectively tracks two energy sums,

$$E_{\text{pot,self}} = \sum_i m_i \, \Phi_{\text{res}}(\mathbf{x}_i), \qquad E_{\text{pot,ext}} = \sum_i m_i \, \Phi_{\text{ext}}(\mathbf{x}_i), \tag{3.10}$$

with Φ_{res} the resolved potential and Φ_{ext} the analytic truncated-NFW contribution. If strict continuity at R_{cut} is desired, the external backend can be shifted by an additive constant so that the potential remains continuous across the cutoff; this constant does not affect the force, but it does change the reported potential energy zero-point.

The external potential may optionally supply a timestep multiplier for outer particles (for example to limit timesteps when the analytic field varies strongly), but by default the domain's existing timestep criteria govern integration.

3.3.7 Centering, cutoff convention and edge cases

The backend uses a single, explicitly provided center for both the NFW potential and the cutoff test. Two configuration choices affect this center: absolute coordinates (`useabspos=1`) or coordinates relative to box origin/periodic convention (`useabspos=0`). The policy is documented at startup and logged.

Particles at exactly $r = R_{\mathrm{cut}}$ are classified as inner ($r \leq R_{\mathrm{cut}}$). This avoids oscillatory reclassification for particles grazing the boundary and is consistent with the mathematical definitions used in the analytic diagnostics.

3.3.8 Serialization, restart safety and diagnostics

All runtime parameters are recorded in the standard external potential state so that restarts reproduce the mask and potential choices. On startup the backend prints a compact diagnostics summary that includes backend name, chosen center, r_{cut}, and whether DM masking is active. In the current repository the masking initialization is performed at fresh start and after restart via a single call to `gravity_sync_truncated_nfw_masses(...)` (see `src/engine.c`), which applies the one-time stored-mass mask described above. To preserve runtime performance this sweep is not re-executed every gravity update; earlier designs that refreshed the stored mass per step (or per drift) were reverted because they introduced significant costs.

Because the modification changes the effective source mass (and therefore total self-gravity source) the global energy bookkeeping routines are instrumented to report the expected non-conservative terms and to show the separation between particle-sourced and analytic-potential contributions to the total potential energy. These diagnostics are intended for validation and are not enabled in tight production runs by default.

The resulting bookkeeping is best viewed as a three part decomposition of the total gravitational response: resolved self-gravity inside the cutoff, analytic external gravity outside the cutoff, and a small non-conservative correction associated with the masked source field. This is why the implementation records both the resolved and external energy channels separately, rather than collapsing them into a single scalar diagnostic.

3.3.9 Foreign particle and MPI consistency

To preserve exact force symmetry across MPI boundaries, the effective mass semantics must be identical for both local and foreign particles. In the current implementation the packed/foreign particle structures carry the stored particle mass directly (i.e. the code exports ‘packed->mass = gp->mass‘, which will be the masked tiny value for outer DM after the startup sweep). This ensures identical multipole and pairwise force construction on all ranks without reconstructing mask factors on the receiver side.

3.3.10 Caveats and limitations

The present implementation aims at a minimal, well defined numerical experiment and therefore imposes several deliberate limitations:

- the cutoff is a hard step; no smooth transition shell is implemented (this can be added later if numerical artefacts at the boundary are observed),
- only dark matter particles are eligible for masking; other particles keep their full source mass, and their coupling to the external potential is unchanged,
- using a fixed centre for the external potential while the particle distribution drifts can create a mismatch; the user must therefore choose the centre carefully (e.g. by using a shrinking-sphere centre estimate before launching a hybrid run),
- energy is not strictly conserved across the split because masked particle mass is removed from the self-gravity source and replaced by an analytic potential,
- masking is performed as a one-time startup/restart sweep that overwrites stored particle masses for outer DM; particles that start outside R_{cut} therefore remain masked even if they later move inward. Restoring full stored mass on re-entry was considered but intentionally not implemented because the required frequent global or per-drift synchronization caused unacceptable runtime overhead.
- for strict energy continuity diagnostics, a constant potential offset at R_{cut} may still be desirable in the external backend (forces are unaffected by this constant, but reported potential energies can shift).

The truncated-NFW implementation is maintained on a SWIFT feature branch named `truncated-NFW` (see `swiftsim/src/potential/truncated_nfw/potential.h` and the associated gravity hooks). The SWIFT repository and the branch are available on GitLab: https://gitlab.cosma.dur.ac.uk/swift/swiftsim.

4 Results and discussion

This chapter presents the numerical results from the simulation campaign and the accompanying sensitivity studies. The baseline evolution of the reference halo and the principal diagnostics are described first; subsequent sections examine robustness to analysis choices (centering and binning), numerical controls (softening, timestepping and solver tolerances), domain size and initial inner slope. Truncated-NFW hybrid experiments are used to isolate the influence of long-range potential contributions.

4.1 Baseline behaviour

The baseline run shows a clear cusp-to-core transition, but the core does not stabilize. Instead, the inner density continues to decline throughout the simulated time span. This is the central empirical result of the thesis. In the broader context of halo evolution, the literature shows that collisionless systems can display a wide range of inner responses to external perturbations and time-dependent forcing, including stochastic heating by substructure and repeated diffusion in energy angular momentum space (Peñarrubia, 2017, 2019). Recent controlled N-body models by Errani et al. (2026) show that collisional star dark matter energy exchange can heat a cusp into a core and that this evolution tends to self-regulate as central dark matter densities drop and binary formation increases, i.e. the contraction/heating rates slow at late times. If that late-time stabilization picture applied directly to our setup, the inner density would be expected to flatten after core formation rather than continue declining. The reference run is therefore notable not because it forms a core, but because the core remains dynamically active rather than approaching a steady inner state.

The result is also instructive from the point of view of numerical convergence. Power et al. (2003) showed that the inner halo is especially sensitive to the interplay between resolution, force softening, and timestep control, while Zhang et al. (2019) later demonstrated that too aggressive a reduction of the softening length can artificially raise the innermost density. The reference simulation does not display a simple monotonic dependence on any one of these parameters, which already suggests that the behaviour is not a trivial one-parameter resolution effect.

To make these statements quantitative a few standard diagnostic definitions are used. All radial profiles are measured in a sequence of spherical shells extending from the halo centre to a maximum radius $r_{\max}$. Rather than adopting uniformly spaced bins, a logarithmically compressed coordinate is employed in order to increase the spatial resolution in the central region while maintaining adequate sampling at large radii. The transformation is defined as

$$u(r) = \ln\left(1 + \frac{r}{r_c}\right), \tag{4.1}$$

where r_c is a characteristic scale radius. The radial bin boundaries are uniformly spaced in the transformed coordinate u, and subsequently mapped back to physical radii through

$$r(u) = r_c\left(e^u - 1\right). \tag{4.2}$$

This construction yields narrow bins near the halo centre and progressively wider bins at larger radii, reducing shot noise while preserving the ability to resolve the inner density structure.

The spherically averaged density profile is obtained by dividing the halo into radial shells and computing the mass enclosed within each shell. If ΔM_k denotes the mass contained in the shell bounded by radii r_k and r_{k+1}, the density is estimated as

$$\rho(r_k) = \frac{\Delta M_k}{\Delta V_k}, \tag{4.3}$$

where the shell volume is

$$\Delta V_k = \frac{4\pi}{3}\left(r_{k+1}^3 - r_k^3\right). \tag{4.4}$$

The resulting profile $\rho(r)$ provides the primary diagnostic for identifying the formation of a central core or other structural changes within the halo.

Profiles are compared to the initial state using ratio curves $\rho(r,t)/\rho(r,t_0)$, which makes secular flattening or depletion immediately visible. Profile plots also include a vertical reference marking the softening-related radius

$$r_{\mathrm{ref}} = 2.8\epsilon, \tag{4.5}$$

where ϵ is the run softening length; this helps separate physically interpretable radii from numerically delicate scales.

In the plot in Figure 4.1 the central density drops rapidly at early times, then continues to drift downward more slowly, without ever forming a stable plateau. The companion plot in Figure 4.2 presents the same evolution and adds the ratio to the initial conditions, which makes the decline easier to read at a glance. In that sense, the evolution resembles a continuing redistribution of mass rather than a completed structural transformation. This is the key point that motivates the rest of the chapter.

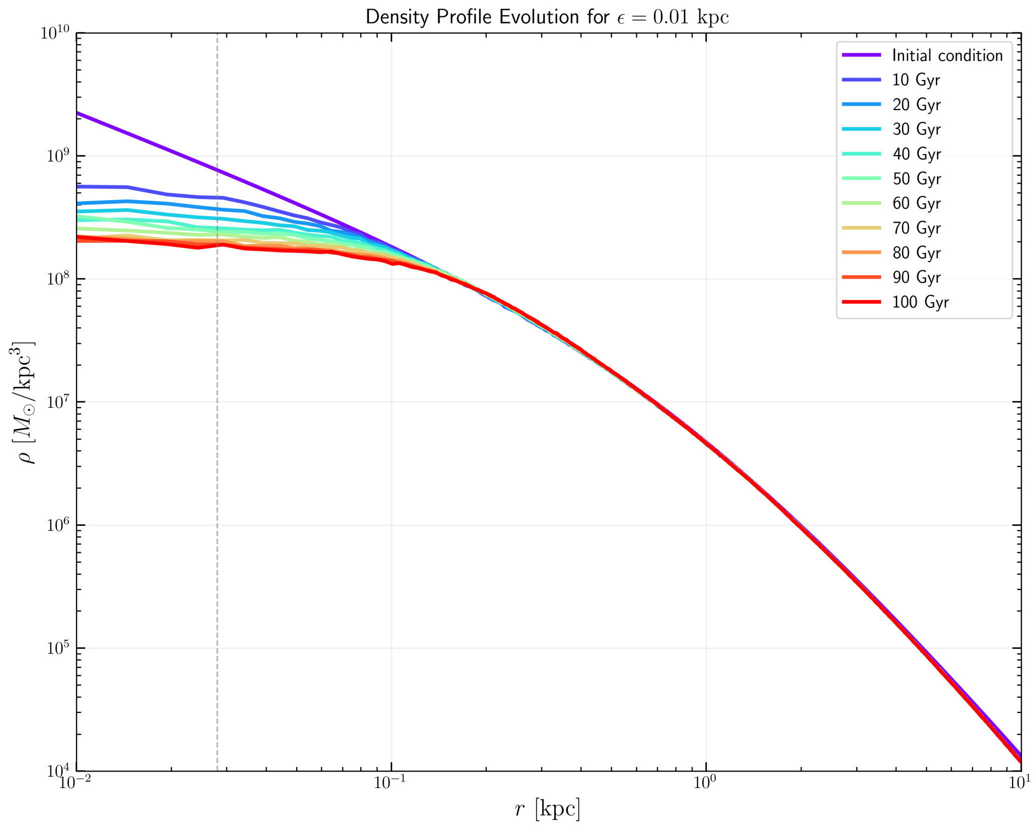


Figure 4.1: **Baseline density-profile evolution.** The central density decreases steadily over time, with the steepest change occurring early and the later evolution approaching a shallower, but still declining, trend. The gradient structure is consistent with a secular redistribution rather than a settled core.

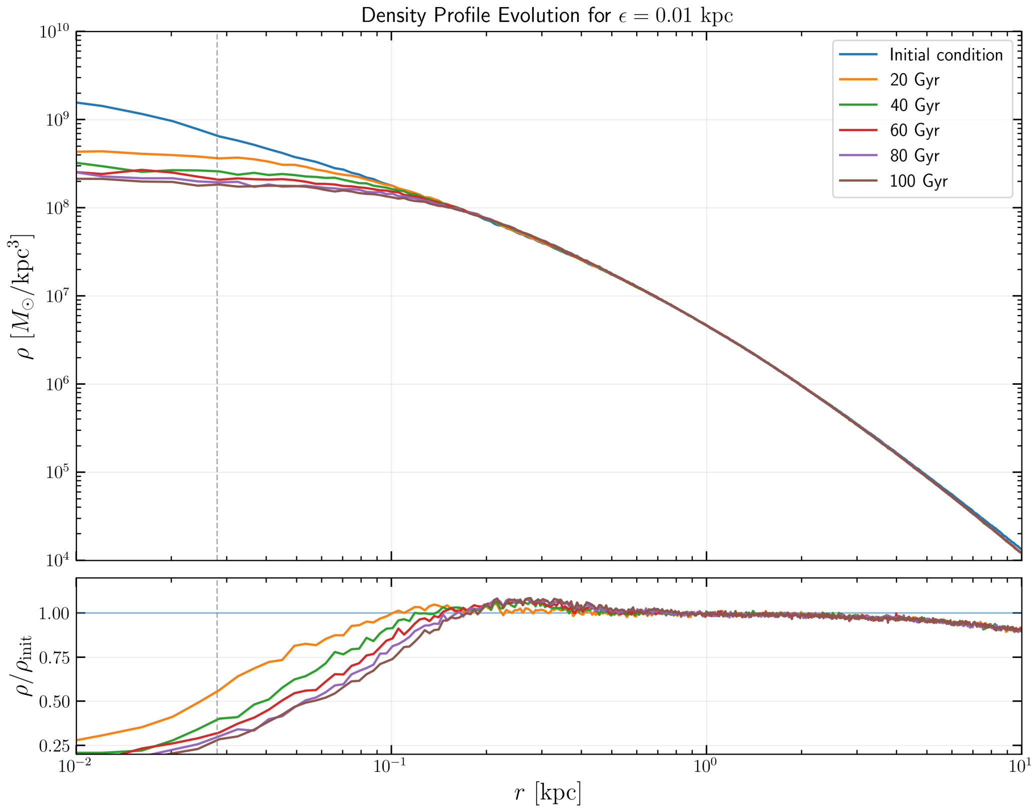


Figure 4.2: **Baseline evolution with ratio to the initial state.** The upper panel shows the density evolution over time. The lower panel shows the density ratio relative to the initial conditions.

4.1.1 Snapshot centering

Centering is handled independently from the density estimation. Each snapshot is recentered using a shrinking-sphere algorithm, which iteratively refines the halo center by repeatedly restricting the particle set to smaller radii around the current estimate. If the particle positions are $\mathbf{x}_i$ and masses are m_i, the mass-weighted centroid of a selected set S_k is

$$\mathbf{x}_k = \mathbf{C}(S_k) = \frac{\sum_{i\in S_k} m_i \mathbf{x}_i}{\sum_{i\in S_k} m_i}, \tag{4.6}$$

with the restricted set updated as

$$S_{k+1} = \{i \in S_k : \|\mathbf{x}_i - \mathbf{x}_k\| < R_k\}, \tag{4.7}$$

$$R_{k+1} = f_s R_k, \tag{4.8}$$

for a shrink factor $0 < f_s < 1$. This method is robust against mild asymmetries and against the gradual drift that can occur in long collisionless runs.

In practice, this removes one of the most obvious numerical artefacts that could otherwise be mistaken for a central density decline. The iteration stops once the centroid shift becomes small, $\|\mathbf{x}_{k+1} - \mathbf{x}_k\| < \delta$, or the retained set becomes too small, $|S_{k+1}| < N_{\min}$. A formal description of the algorithm and the exact analysis implementation are given in Appendix A.

4.1.2 Kinematic heating analysis

In an ideal collisionless dark matter halo, particles evolve within a smooth gravitational potential and should not exchange significant amounts of energy through direct encounters. However, in an N-body simulation the finite number of particles introduces discreteness noise, allowing weak particle-particle interactions to accumulate over time.

The hallmark of this process is a gradual and approximately linear increase in the velocity dispersion or in the anisotropy parameter, β, throughout the entire simulation, even over timescales as long as 100 Gyr. This behavior indicates that the system is being artificially heated by numerical effects rather than evolving due to physical mechanisms.

The consequences of two-body relaxation can be significant. Continuous energy exchange between particles can alter orbital distributions, modify the velocity structure of the halo, and potentially generate artificial core growth or changes in the density profile. Consequently, detecting a long-term monotonic increase in velocity dispersion or anisotropy serves as strong evidence that numerical relaxation is influencing the simulation results.

To quantify whether the apparent core evolution is driven by genuine mass redistribution or by numerical/physical heating, the second moments of the velocity distribution in spherical shells are examined. For each snapshot, the halo is first recentered and the bulk velocity is subtracted from all particles. For a particle i at position $\boldsymbol{x}_i$, its distance from the center is defined as,

$$r_i = |\boldsymbol{x}_i|, \tag{4.9}$$

and the corresponding unit radial vector,

$$\hat{\boldsymbol{r}}_i = \frac{\boldsymbol{x}_i}{r_i}. \tag{4.10}$$

The particle velocity relative to the bulk motion is denoted by $\boldsymbol{v}_i$.

The radial component of the velocity is then

$$v_{r,i} = \boldsymbol{v}_i \cdot \hat{\boldsymbol{r}}_i, \tag{4.11}$$

while the tangential velocity is obtained from the decomposition

$$v_{t,i}^2 = |\boldsymbol{v}_i|^2 - v_{r,i}^2. \tag{4.12}$$

The particles are grouped into spherical radial bins centered at radius r_k. In each bin, the velocity dispersions as the unbiased variances of the corresponding velocity components are computed:

$$\sigma_r^2(r_k) = \langle v_r^2 \rangle_k - \langle v_r \rangle_k^2, \tag{4.13}$$

$$\sigma_t^2(r_k) = \langle v_t^2 \rangle_k - \langle v_t \rangle_k^2, \tag{4.14}$$

where $\langle \cdot \rangle_k$ denotes an average over all particles in the k-th shell. Equivalently, one may compute the total velocity variance in the shell and subtract the radial contribution:

$$\sigma_t^2(r_k) = \sigma_{\mathrm{tot}}^2(r_k) - \sigma_r^2(r_k), \tag{4.15}$$

with

$$\sigma_{\mathrm{tot}}^2(r_k) = \sigma_x^2(r_k) + \sigma_y^2(r_k) + \sigma_z^2(r_k). \tag{4.16}$$

From these quantities, the velocity anisotropy parameter is defined as

$$\beta(r_k) = 1 - \frac{\sigma_t^2(r_k)}{2\sigma_r^2(r_k)}. \tag{4.17}$$

An isotropic velocity distribution corresponds to $\beta = 0$, while $\beta > 0$ indicates radially biased orbits. In the context of core evolution, a sustained increase in σ_r and/or a systematic rise in β in the innermost shells would signal kinematic heating rather than simple adiabatic expansion.

To further characterize the system's kinematic state, the one-dimensional velocity dispersion is computed. The total one-dimensional velocity dispersion, $\sigma_{1\mathrm{D}}(r)$, is defined via its radial (σ_r) and tangential (σ_t) components as:

$$\sigma_{1\mathrm{D}}(r) = \sqrt{\frac{\sigma_r^2(r) + \sigma_t^2(r)}{3}} = \sqrt{\frac{\sigma_{\mathrm{tot}}^2(r)}{3}}. \tag{4.18}$$

Because the kinematic profiles are measured from finite particle numbers, the raw binned quantities exhibit statistical noise, especially at small radii. To improve visual clarity while preserving trends, the profiles are optionally smoothed with a one-dimensional Gaussian kernel in radius. For a generic profile $X(r_k)$, the smoothed profile is defined as

$$\widetilde{X}(r_k) = \frac{\sum_j X(r_j) \exp\left[-\frac{(r_k - r_j)^2}{2h^2}\right]}{\sum_j \exp\left[-\frac{(r_k - r_j)^2}{2h^2}\right]}, \tag{4.19}$$

where h is the smoothing length. In practice, h is chosen to be comparable to one radial bin width, so that the filter suppresses shot noise without erasing genuine radial trends. The same procedure is applied consistently to $\sigma_r(r)$, $\sigma_t(r)$, and $\beta(r)$ when comparing different epochs.

For a collisionless stellar system in steady-state, spherical hydrostatic equilibrium, the velocity dispersion profiles and the spatial density distribution $\rho(r)$ must satisfy the spherical Jeans equation:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left[\rho(r)\sigma_r^2(r)\right] + \frac{2\beta(r)}{r}\rho(r)\sigma_r^2(r) = -\rho(r)\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{GM(<r)\rho(r)}{r^2}, \tag{4.20}$$

where $\beta(r)$ is the stellar anisotropy parameter defined previously, $\Phi(r)$ is the global gravitational potential, and $M(<r)$ represents the cumulative halo mass contained within a sphere of radius r.

The above diagnostics are computed at multiple simulation times, and the resulting profiles $\sigma_{1\mathrm{D}}(r)$ and $\beta(r,t)$ are compared between the initial state and late-time snapshots in Figure 4.3.

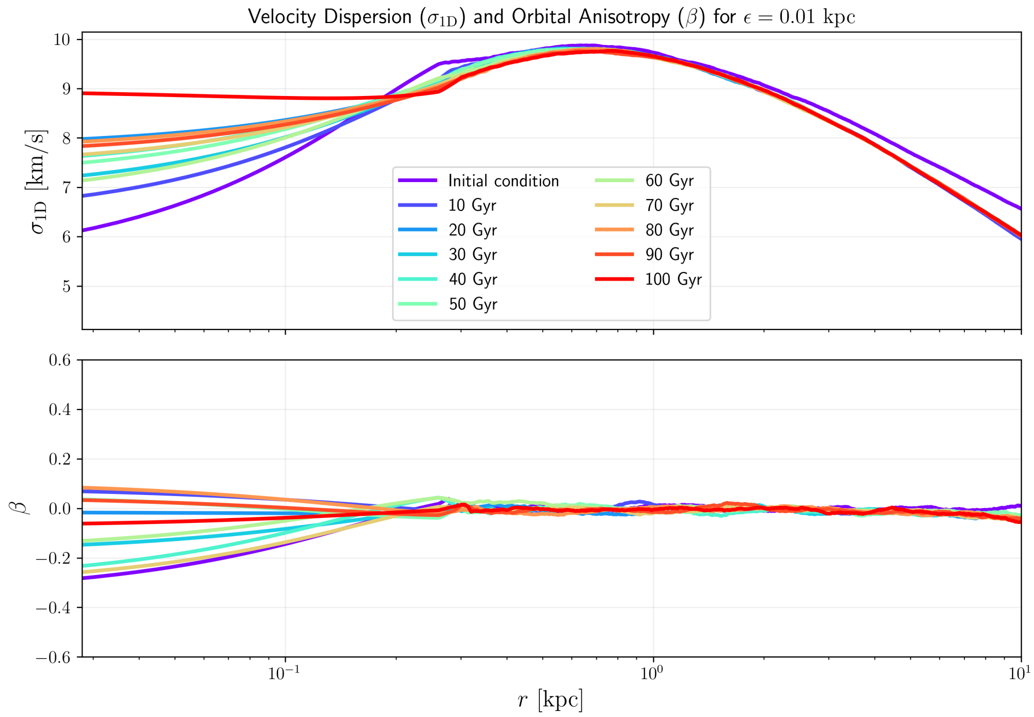


Figure 4.3: Velocity dispersion and orbital anisotropy evolution up to 100 Gyr for a softening length value of $\epsilon = 0.01$ kpc.

The evolution of the isolated halo provides clear evidence for numerical core heating over the full 100 Gyr integration. In the inner region, $r \lesssim 0.1$ kpc, the one-dimensional velocity dispersion, $\sigma_{1\mathrm{D}}$, increases continuously from its initial value of approximately 6 km s^{-1} to nearly 9 km s^{-1} by the end of the run. This trend is physically significant because an expanding, declining density system would be expected to cool adiabatically; instead, the halo becomes kinematically hotter as the central density decreases. The simultaneous drop in density and rise in $\sigma_{1\mathrm{D}}$ therefore rules out adiabatic expansion and strongly indicates non-adiabatic heating. For an isolated system, this behavior is most naturally interpreted as numerical heating driven by discreteness noise and two-body relaxation. In this interpretation, spurious energy injection gradually puffs up the central region, lowering the density while increasing the random particle motions.

In contrast to the steady chronological rise observed in the velocity dispersion, the orbital anisotropy parameter, β, does not exhibit a directional evolution over time. The initial condition begins at a non-zero value, and subsequent snapshots appear heavily intermingled, fluctuating without a clear chronological sequence in the inner regions ($r \lesssim 0.1$ kpc). This lack of a coherent temporal trend indicates that while numerical heating continuously injects kinetic energy into the halo core, it does not alter the underlying orbital geometry of the system over 100 Gyr. Instead, chaotic nature of the β profiles highlights that the inner velocity architecture is dominated by snapshot-to-snapshot stochastic noise. This high variability is likely driven by low particle numbers in the innermost log-spaced bins, or by minor physical sloshing of the halo core relative to the calculated center of mass, which dynamically redistributes the measured radial and tangential velocity components from one snapshot to the next.

4.2 Box size

The role of the numerical domain was tested explicitly by repeating the reference experiment in three increasingly larger simulation volumes while holding all physical and numerical parameters fixed (identical particle realization, softening, timestepping rules and gravity solver settings). The objective was to determine whether finite domain effects or boundary condition artefacts can explain the persistent central density decline.

Figure 4.4 summarises the outcome. The central density diagnostic used throughout this thesis shows the same secular decline in each volume. Enlarging the domain alters the quantitative rate of depletion by a modest amount, but it does not halt or reverse the process; the qualitative behaviour is invariant to the choice of box size.

There are two simple physical interpretations of this result. First, long-range mass beyond the immediate inner region changes the background potential and can therefore modify rates (Binney and Tremaine, 2008). Second, if the mechanism were purely an artefact of a truncated domain (for example a spurious boundary tide or reflection), one would expect the effect to vanish as the box is enlarged; the absence of such a disappearance argues against this idea. These conclusions are in line with standard convergence diagnostics for collisionless codes (Power et al., 2003) and with the expectation that stochastic long-range perturbations modulate rates rather than create new inner dynamics (Peñarrubia, 2017, 2019).

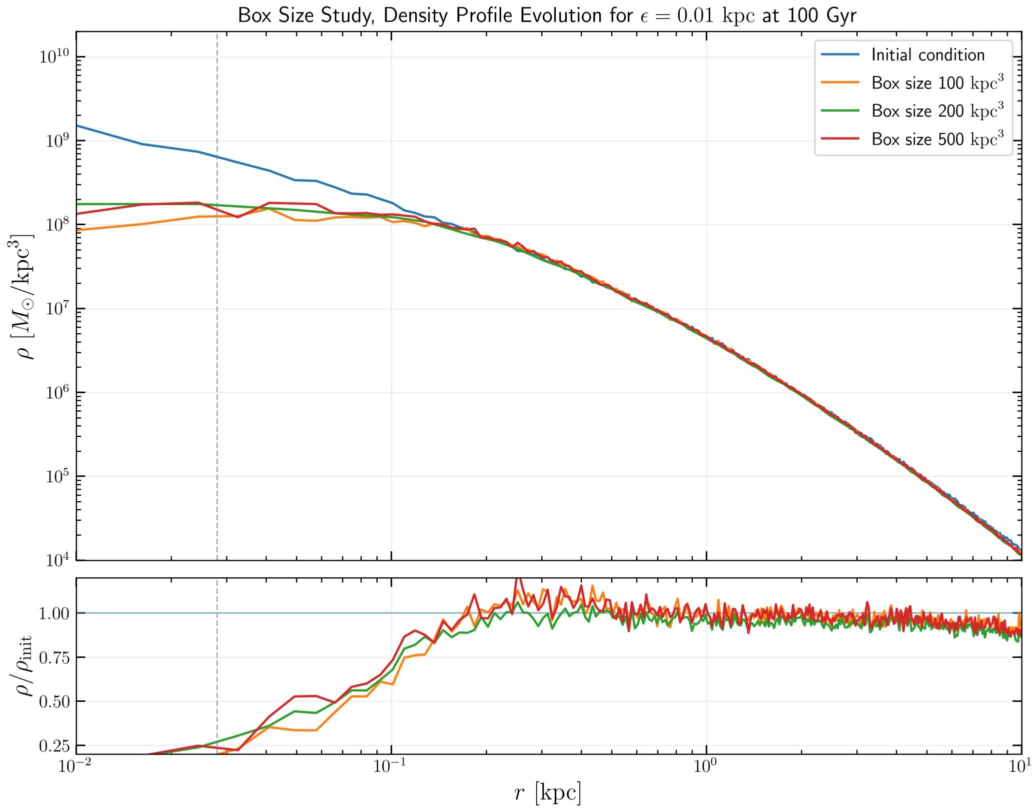


Figure 4.4: **Box-size study.** Comparison of the three tested simulation volumes (small, intermediate, large). The central-density diagnostic (inner radial bin) shows the same secular decline across volumes; changing the box size modifies the depletion rate only modestly.

4.3 Initial halo structure

The comparison between cuspy and cored initial conditions is particularly informative because it distinguishes between a cusp specific instability and a more general process. The core run still exhibits a decline of the central density, but the amplitude is smaller than in the cuspy case. Shown in Figures 4.5 and 4.6, this is a strong result because it means the mechanism is not limited to the special case of an initially steep central profile.

In physical terms, this suggests that the system is sensitive to the presence of a central potential well, but not exclusively to its initial logarithmic slope. A cored halo has less extreme inner structure and therefore reacts less strongly, yet the evolution remains qualitatively the same. This is particularly important in light of the Errani et al. (2026) result that cusp-heating evolution should weaken once the core is established and central dark matter densities are reduced. In that scenario, a pre-existing core should not continue to lose central density over long timescales; however, this behaviour is observed in the simulations. That mismatch reinforces the conclusion that the persistent decline cannot be explained as a simple cusp-to-core transient and is instead tied to a broader dynamical mechanism.

The literature on heating caused by substructure is again relevant here. Stochastic forcing does not require a cusp; it requires a fluctuating potential and a population of perturbers that can exchange energy with the central region (Peñarrubia, 2017). The persistence of the effect in the core run is therefore compatible with a non-local heating picture and inconsistent with the idea that the problem is restricted to cuspy initial conditions.

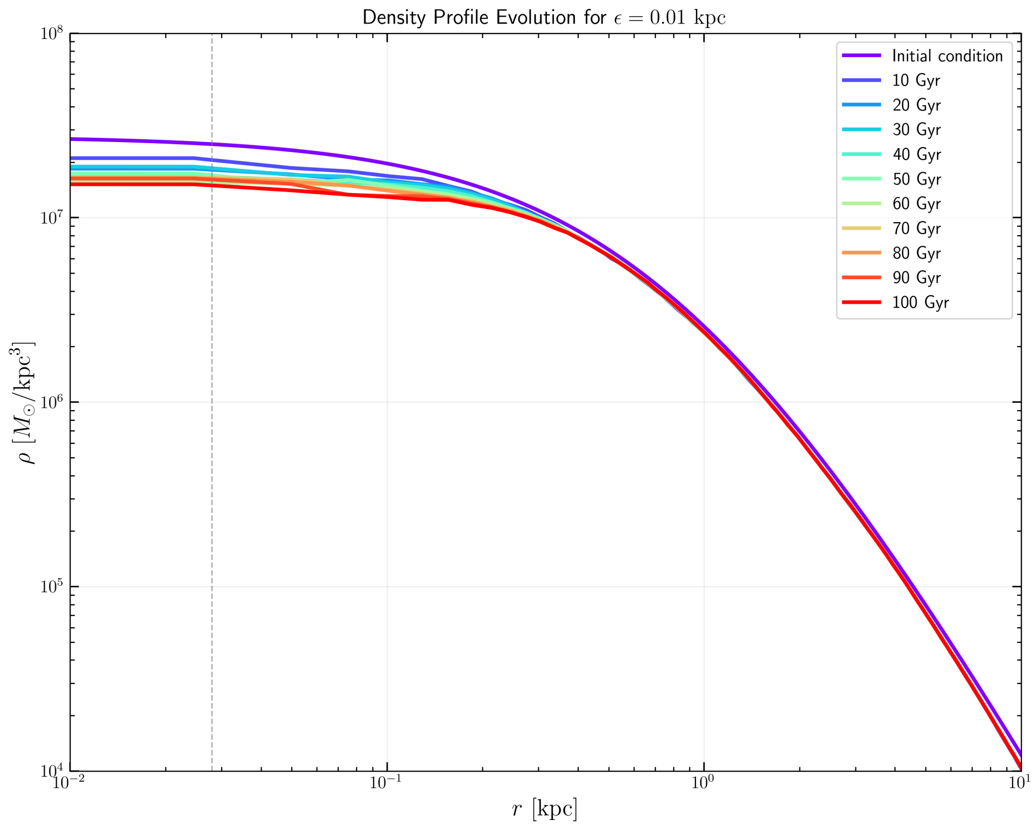


Figure 4.5: **Core initial condition density-profile evolution.** The cored halo also shows a decline in central density, although with a smaller amplitude than the cuspy case. This directly contradicts a simple expectation of core formation followed by stabilization.

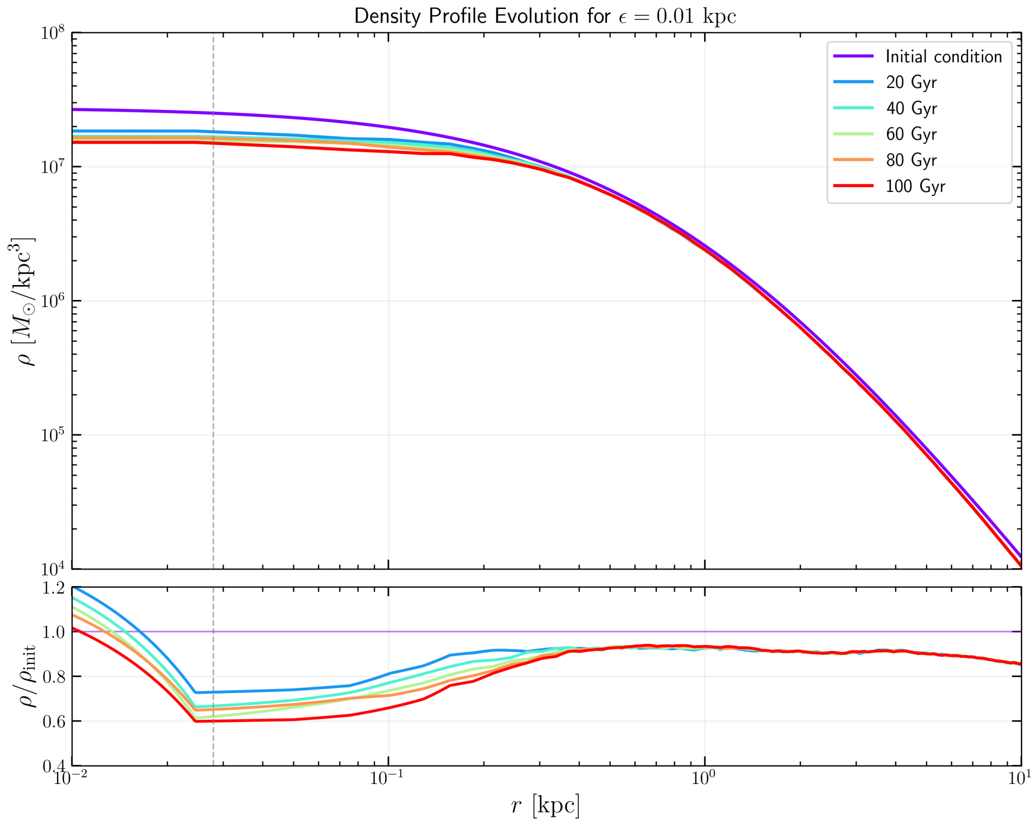


Figure 4.6: **Core initial condition evolution with ratio to the initial state.** The upper panel shows the density evolution over time. The lower panel shows the density ratio relative to the initial conditions.

4.4 Gravitational softening

Softening is the most obvious numerical parameter to suspect in any study of the inner halo, because it directly regularizes short-range interactions. The literature makes clear, however, that the role of softening is subtle: too much softening suppresses small scale structure, while too little can artificially increase the density in the innermost region (Power et al., 2003; Zhang et al., 2019). In the convergence analysis of Zhang et al. (2019), the optimal softening length is the one that balances force bias against discreteness noise; below that scale, the central profile can become biased high because close encounters are less regularized and the inner region responds more strongly to numerical heating rather than to a genuine improvement in resolution. The present results fit that picture in the sense that the softening length changes the detailed rate of evolution, but not the existence of the decline itself. The tested softening values are $\epsilon = 1, 5, 10, 50$, and 100 pc; shown in Figure 4.7 are the 50 Gyr and 100 Gyr snapshots.

This is an important negative result. If softening were the primary cause, the central density evolution should have shown a qualitatively different response when the softening length was varied.

Instead, the trend persists across the tested range. The smallest tested softening shows an increase in density as one approaches the center, but that should not be interpreted as a real improvement of the physical problem. The more likely explanation is that the reduced softening has allowed stronger numerical heating and a slightly more concentrated inner profile, which is exactly the kind of bias that an under-softened calculation can introduce (Power et al., 2003; Zhang et al., 2019).

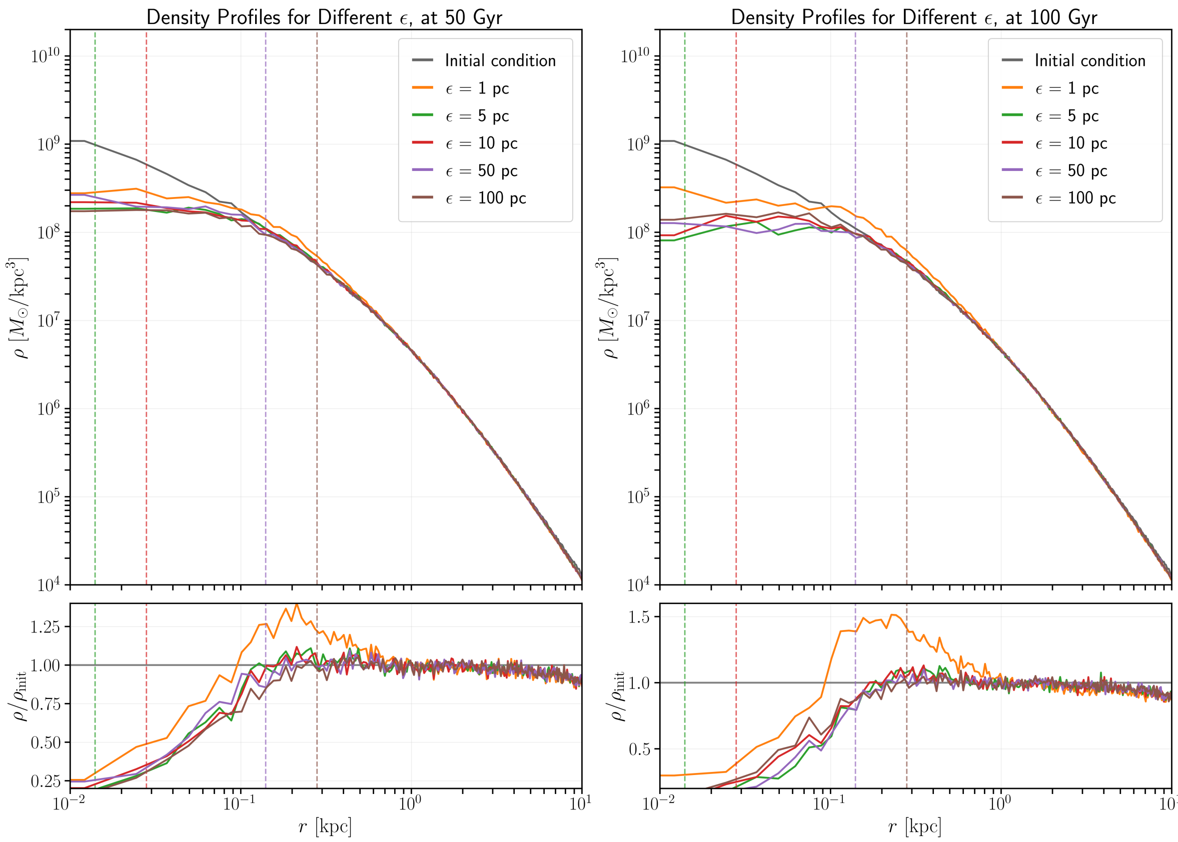


Figure 4.7: **Softening study.** Comparison of the same set of tested softening lengths (ϵ = 1, 5, 10, 50, and 100 pc) at 50 Gyr (left) and 100 Gyr (right).

4.5 Self-gravity settings

The solver parameter study addresses whether the observed decline could be the cumulative effect of force approximation or integration errors. This is a natural concern in gravity solvers where the acceptance criterion and multipole tolerances determine how accurately long range interactions are represented. In principle, a slightly looser or tighter force approximation can shift the evolution of a halo even when the initial conditions are identical.

The present study varies four key control parameters of the self-gravity solver: timestep factor η, Multipole Acceptance Criterion (MAC), geometric opening angle θ_{cr} and FMM tolerance ϵ_{fmm}.

The parameter range explored here is broad enough to reveal systematic sensitivity if it exists.

The timestep factor was varied by roughly an order of magnitude, the MAC was changed between geometric and adaptive formulations, θ_{cr} was tightened in the geometric case, and ϵ_{fmm} was varied over two to three orders of magnitude. In a purely numerical explanation, one would expect at least one of these settings to have a decisive effect on the inner profile. Results are represented in Figure 4.8.

Table 4.1: Summary of self-gravity solver parameter values in the study. Each row lists the parameter, its baseline value (used in the reference run), and the test values explored to assess sensitivity.

Parameter	**Baseline**	**Test values**		
η (timestep)	0.002	0.0002	0.02	—
MAC	geometric	adaptive	—	—
θ_{cr} (geometric only)	0.7	0.6	0.5	—
ϵ_{fmm} (FMM tolerance)	0.001	0.0001	0.01	0.1

4.5.1 Timestep study

The timestep factor η determines the frequency of gravity and kinematics updates. Increasing η allows larger timesteps and reduces the number of force evaluations per unit time; decreasing η enforces finer temporal sampling and can improve phase-space accuracy. The baseline value of $\eta = 0.002$ is typical for high-resolution dark matter studies, but it is instructive to test both more conservative ($\eta = 0.0002$, roughly $10\times$ finer) and more permissive ($\eta = 0.02$, roughly $10\times$ coarser) variants.

If timestep error were the main driver of central decline, the coarser run should exhibit notably faster or more severe depletion, while the finer run might show stabilization. That is not observed.

4.5.2 Multipole acceptance criterion study

The MAC choice determines when the force from a distant cell can be approximated by its multipole moments rather than evaluated exactly from individual particles. The *geometric* MAC uses the cell size and distance; the *adaptive* MAC adjusts the criterion based on the local particle distribution. Geometric MAC is faster but less accurate for clumpy structures.

If MAC were the primary sensitivity, changing between geometric and adaptive should produce a qualitatively different central decline rate. The runs tested here show no such distinction.

4.5.3 Geometric opening angle study

For the geometric MAC, a smaller opening angle θ_{cr} means stricter multipole approximation. More cell pairs are forced into exact evaluation rather than multipole summation. The baseline value $\theta_{cr} = 0.7$ is a standard choice; the smaller test values represent a more stringent criterion and therefore higher force accuracy.

Lowering θ_{cr} should increase computational cost but also increase force precision. If force approximation error were limiting the simulation, tightening θ_{cr} should reduce or delay the central decline. The observed effect is marginal.

4.5.4 FMM tolerance study

The Fast Multipole Method tolerance ϵ_{fmm} is a global precision parameter for the multipole expansion. The baseline value is $\epsilon_{\mathrm{fmm}} = 0.001$, and the test suite includes both looser values (0.01 and 0.1, representing $10\times$ and $100\times$ relaxation) and a tighter value (0.0001, representing $10\times$ tightening). This range is designed to reveal whether force approximation is a controlling factor.

Some runs with modified ϵ_{fmm} exhibit slightly improved early time behaviour, particularly the tighter tolerance, but even those runs exhibit long-term central decline. The change is not enough to alter the qualitative long term trend.

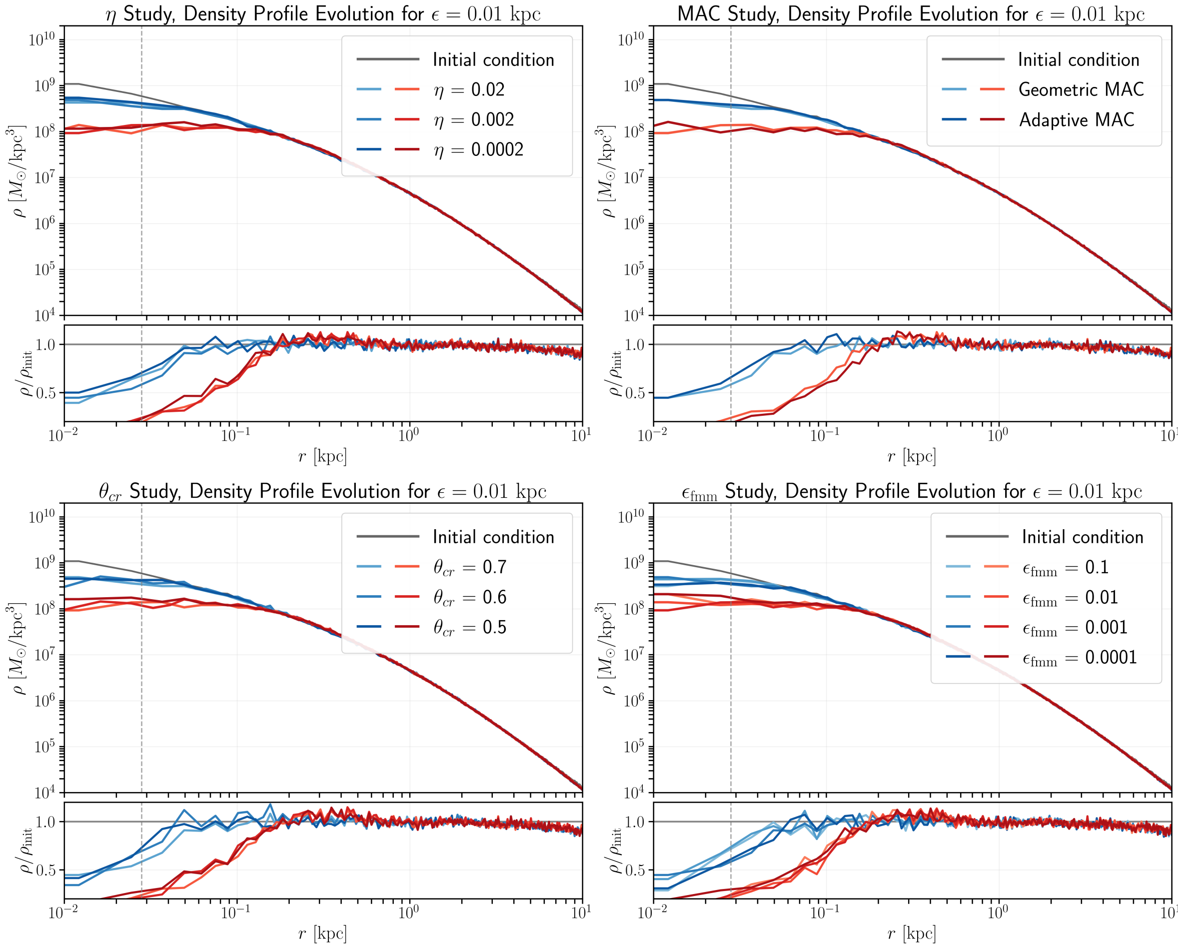


Figure 4.8: **Self-gravity parameter study.** Panels show the central-density evolution for the tested solver settings: top-left – timestep factor η; top-right – MAC choice; bottom-left – geometric opening angle $\theta_{\rm cr}$; bottom-right – FMM tolerance $\epsilon_{\rm fmm}$. In each panel blue curves correspond to 10 Gyr and red curves to 100 Gyr.

The central density evolution remains broadly similar across all tested settings. This is consistent with the general lesson from convergence studies, changing a gravity control parameter may reduce error bars or shift the onset of bias, but it does not necessarily eliminate the physical or numerical process that dominates the evolution (Power et al., 2003; Zhang et al., 2019).

Therefore, the solver settings are not the principal explanation; their influence is secondary. A solver parameter would need to have produced a qualitatively different central decline (either eliminating it or dramatically accelerating it) to be considered a dominant factor. None of the parameters tested here achieve that threshold.

4.6 Outer halo influence on inner potential

The central density decline observed in the baseline simulation could, in principle, arise from dynamical interactions between the inner halo and the outer shells. To test this hypothesis, an analytic shell decomposition study using an NFW model (Navarro et al., 1996) is performed to quantify how much of the inner potential depth comes from material at different radii. This analysis then motivates a hybrid numerical experiment: if outer shells are dynamically important, replacing them with an analytic potential while keeping the inner region as particles should produce a different evolution.

4.6.1 Shell diagnostics and decomposition

The halo is partitioned into concentric spherical shells and each shell's mass contribution and its influence on the central potential are computed. For shell i with mass ΔM_i and representative radius $\bar{r}_i$, the shell contributes approximately

$$\Phi_i(0) \approx -G\frac{\Delta M_i}{\bar{r}_i} \tag{4.21}$$

to the potential at the origin. By contrast, the radial force at a radius r depends only on the enclosed mass $M(< r)$ by the shell theorem, so outer shells do not change the local force gradient but they do deepen the potential well and therefore the escape speed scale. The shell diagnostics therefore track both $M(< r)$ and the cumulative potential contribution which is useful to quantify non-local coupling between outer mass and inner dynamics.

For an NFW profile with parameters $\rho_s = 10^7\, M_\odot/\mathrm{kpc}^3$, $r_s = 10\,\mathrm{kpc}$, $r_{\mathrm{vir}} = 100\,\mathrm{kpc}$, the analysis computes: (i) the enclosed mass $M(< r)$ and shell mass distribution, (ii) the contribution of each shell to the potential depth at the center, and (iii) the fraction of potential at arbitrary probe radii originating from material outside a chosen cutoff radius. Key diagnostics and results are are given in Appendix B.

The analytic shell study establishes that the outer halo is not dynamically negligible, it contributes substantially to the potential structure experienced by inner particles, even if it does not directly modify the local force field. This has two important consequences:

1. **Motivation for the hybrid model**: A simulation in which the inner region ($r < r_{\mathrm{cut}}$) evolves with particles while the outer region ($r > r_{\mathrm{cut}}$) is represented by an analytic potential can only reproduce the true inner evolution if the outer potential is represented accurately.
2. **Tests of shell coupling mechanisms**: If the persistent central density decline arises from global halo coupling, where the outer shells modulate the potential experienced by inner particles, then a hybrid model should produce a noticeably different evolution. Conversely, if the decline persists even in the hybrid setup, the mechanism is either purely inner region dynamics or depends on structures (e.g., substructure, asymmetries) not captured by a smooth

analytic outer potential.

This is addressed by the hybrid model simulations discussed in the next section.

4.7 Truncated NFW

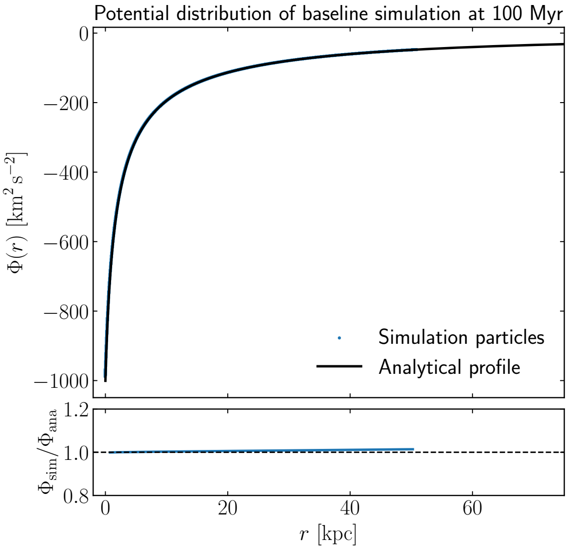
This section introduces the truncated-NFW hybrid experiments used to isolate the role of long range potential on the inner halo evolution. The implementation is first validated with a small suite of diagnostics to ensure the hybrid coupling behaves as intended. After validation, a systematic study of the truncation radius R_{cut} and its impact on the inner density evolution is presented; the results of that Rcut parameter sweep are discussed in the following subsections.

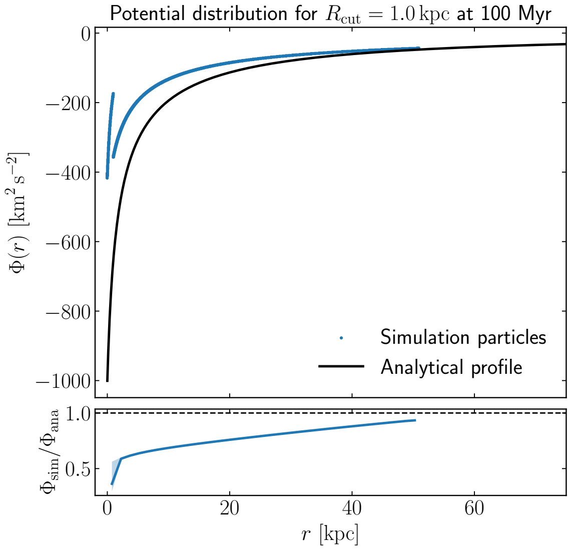
4.7.1 Mass and potential structure

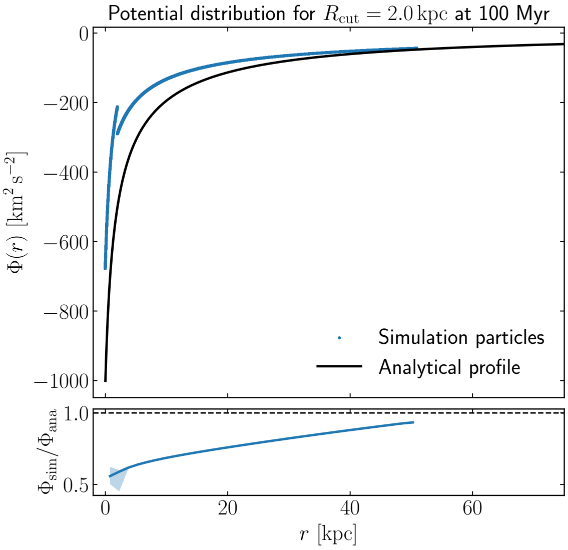
As an initial validation of the truncation procedure, the halo mass partition is verified to be correct. For all values of the truncation radius, the measured mass enclosed within R_{cut} agrees with the expected particle mass, while the remaining halo mass is consistently represented by the analytic potential.

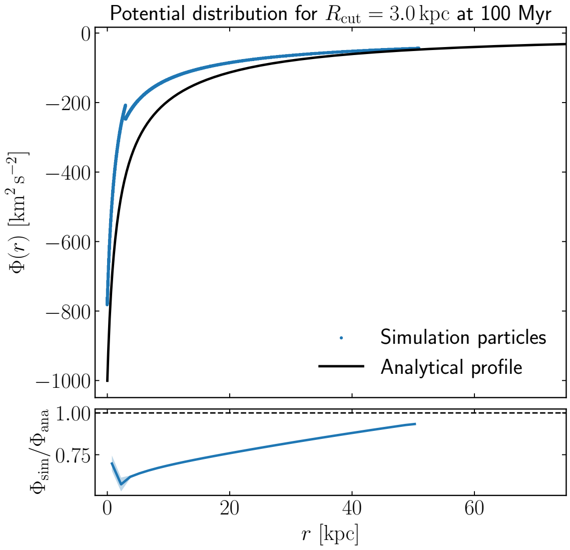
A second validation concerns the resulting gravitational potential (Figure 4.9). The potential distributions obtained for the full halo and for a sequence of truncated models with $R_{\text{cut}} = 50, 25, 10, 5, 3, 2$, and 1 kpc show the expected behaviour. As the truncation radius is reduced, the contribution from distant particles is progressively removed and replaced by the analytic component. The sequence of panels clearly illustrate how the potential smoothly approaches the analytic form for sufficiently large cutoffs.

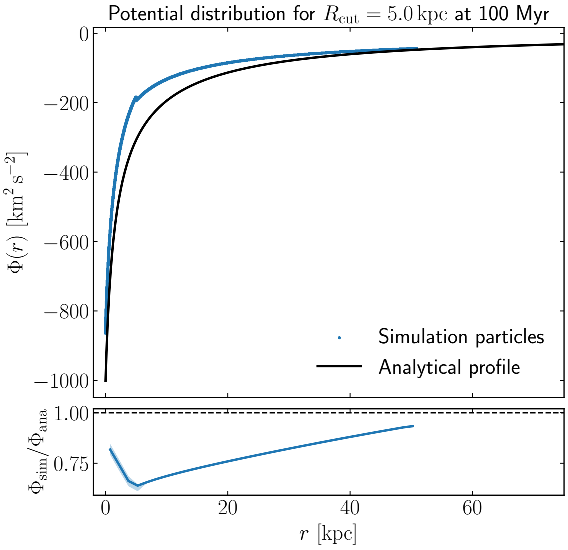
4.7 Truncated NFW

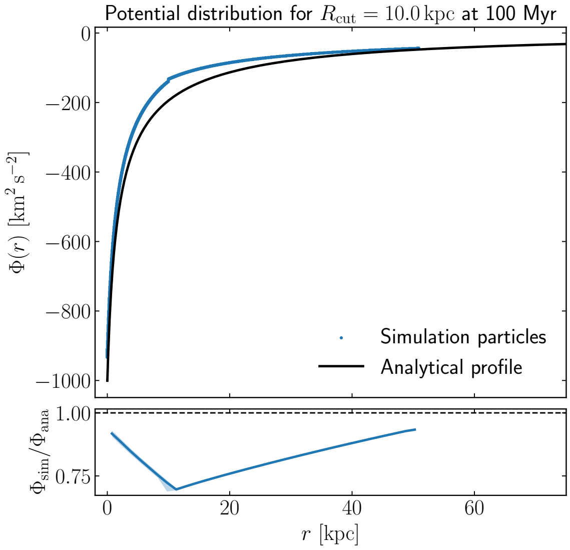












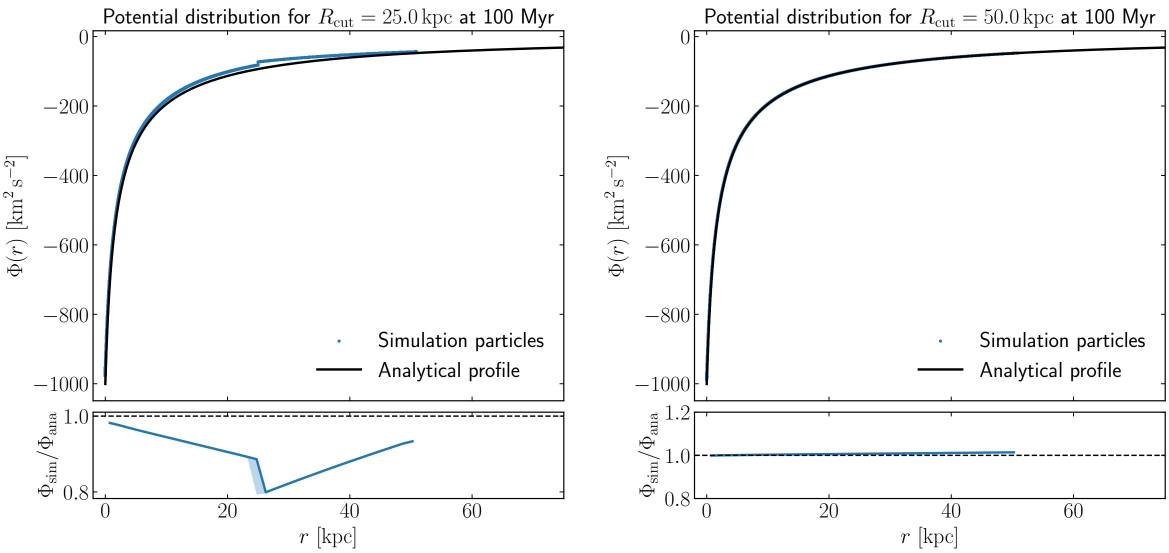


Figure 4.9: **Gravitational potential distributions under baseline and truncated interaction ranges.** Left to right, top to bottom: vanilla *SWIFT* (no truncation), and truncated implementations with $R_{\rm cut} = 50$, 25, 10, 5, 3, 2, and 1 kpc.

4.7.2 Orbits crossing the truncation boundary

Selected test particles that cross the truncation boundary were integrated to check for energy or angular momentum transfer caused by the hybrid partition. An example orbit is shown in Figure 4.10 and remains regular and physically plausible, indicating the hybrid coupling does not introduce obvious integration artefacts on the tested timescales.

We identify particles that move from the inner region, $r < R_{\rm cut}$, to the outer region, $r > R_{\rm cut}$, between two consecutive simulation snapshots. Let the snapshots be indexed by $i = 0, \ldots, N_{\rm snap} - 1$, with snapshot time t_i. For each particle p present in a given snapshot, we denote its position by

$$\mathbf{x}_p^{(i)} = \left(x_p^{(i)}, y_p^{(i)}, z_p^{(i)}\right), \tag{4.22}$$

and define the radius relative to the galaxy centre $\mathbf{x}_{\rm c}$ as

$$r_p^{(i)} = \left\| \mathbf{x}_p^{(i)} - \mathbf{x}_{\rm c} \right\| = \sqrt{(x_p^{(i)} - x_{\rm c})^2 + (y_p^{(i)} - y_{\rm c})^2 + (z_p^{(i)} - z_{\rm c})^2}. \tag{4.23}$$

We define an indicator function for whether particle p is inside the cut radius at snapshot i:

$$I_p^{(i)} = \begin{cases} 1, & r_p^{(i)} < R_{\rm cut}, \\ 0, & r_p^{(i)} \geq R_{\rm cut}. \end{cases} \tag{4.24}$$

A particle is flagged as a crossing particle if it satisfies an outward transition between two consecutive snapshots,

$$I_p^{(i-1)} = 1 \qquad \text{and} \qquad I_p^{(i)} = 0, \tag{4.25}$$

for at least one snapshot pair $(i-1, i)$. In other words, we detect the first discrete time interval in which the particle changes from the inner to the outer regime.

To reduce contamination from particles that are not close to the transition boundary, we optionally apply a radial preselection window. Specifically, a particle is only considered a valid crosser if, at the previous snapshot, its radius lies in the interval

$$R_{\rm cut} - \alpha \le r_p^{(i-1)} < R_{\rm cut}, \tag{4.26}$$

where α is a user-defined thickness parameter. This ensures that only particles that start close to the boundary are included in the crossing sample.

The detection criterion can therefore be written as

$$\mathscr{C}_p^{(i)} = \begin{cases} 1, & \text{if } r_p^{(i-1)} \in [R_{\rm cut} - \alpha, R_{\rm cut}) \text{ and } r_p^{(i-1)} < R_{\rm cut} \le r_p^{(i)}, \\ 0, & \text{otherwise.} \end{cases} \tag{4.27}$$

When $\mathscr{C}_p^{(i)} = 1$, particle p is recorded as a crossing particle and the snapshot pair $(i-1, i)$ is stored together with the corresponding times (t_{i-1}, t_i).

Since the snapshots are discrete in time, the exact crossing time is not known from the data used here. The implementation therefore identifies the crossing within the interval $[t_{i-1}, t_i]$ rather than interpolating to a continuous crossing time. If needed, a linear estimate of the crossing time could be defined as

$$t_{\rm cross} \approx t_{i-1} + \frac{R_{\rm cut} - r_p^{(i-1)}}{r_p^{(i)} - r_p^{(i-1)}} \, (t_i - t_{i-1}), \tag{4.28}$$

but this interpolation is not required for the present selection procedure.

Once the set of crossing particle IDs has been determined, we perform a second pass over all snapshots and extract the full time series for each selected particle. For each recorded particle p and snapshot i, we store

$$\left(t_i,\, x_p^{(i)},\, y_p^{(i)},\, z_p^{(i)},\, r_p^{(i)},\, v_{x,p}^{(i)},\, v_{y,p}^{(i)},\, v_{z,p}^{(i)},\, |\mathbf{v}_p^{(i)}|,\, \Phi_p^{(i)}\right), \tag{4.29}$$

where

$$|\mathbf{v}_p^{(i)}| = \sqrt{(v_{x,p}^{(i)})^2 + (v_{y,p}^{(i)})^2 + (v_{z,p}^{(i)})^2}, \tag{4.30}$$

and $\Phi_p^{(i)}$ is the particle potential when available in the snapshot output.

This produces a compact particle-level data set containing only trajectories relevant to the boundary-crossing test. The resulting time series are then used for post-processing diagnostics, including trajectory visualisation, radius evolution, velocity evolution, and potential checks around $R_{\rm cut}$.

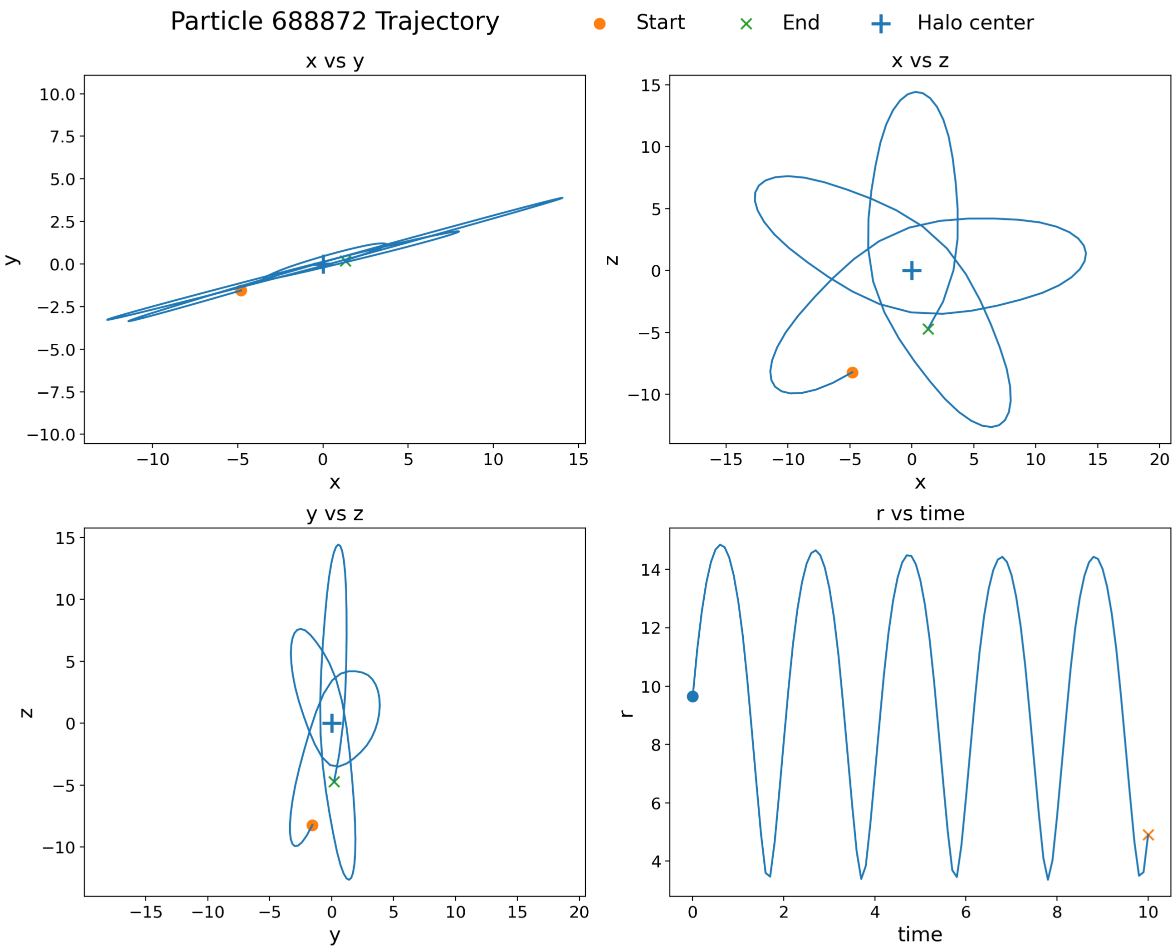


Figure 4.10: **Example orbits crossing the truncation boundary.** Selected test particles that cross R_{cut} are shown to verify the hybrid treatment does not introduce noticeable integration artefacts on the tested timescale.

More orbit plots can be found in the annex Section C. A larger number of orbits is summarized in the following plot. In all cases, the orbits remain regular and show no inconsistencies when crossing the truncation boundary.

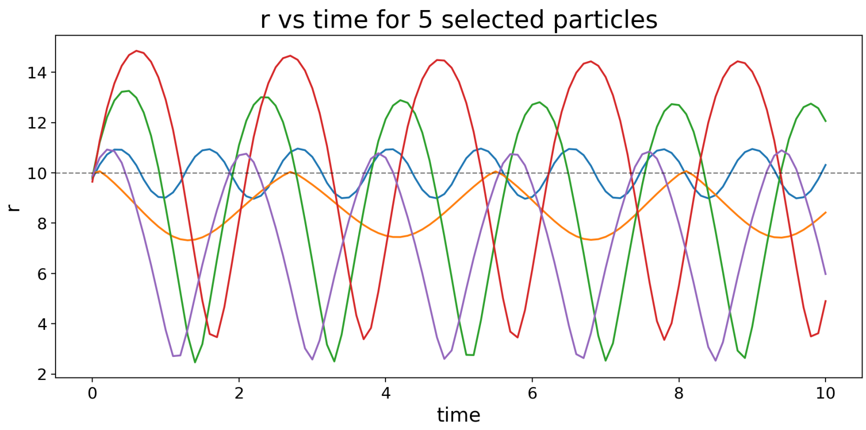


Figure 4.11: **Summary of selected orbits**

In summary, the validation diagnostics show that (i) mass is correctly split by the truncation routine, (ii) the inner potential follows the expected NFW-like shape and tracks the baseline within the inner few kiloparsecs (Figure 4.9), and (iii) orbits are not affected by the truncation (Figure 4.11). These outcomes indicate that the truncated interaction implementation behaves as intended for the described experiments.

4.7.3 Density profile

A complete 100 Gyr run of the density evolution for Rcut = 10 kpcs is shown here. The plot shows an evolution parallel to the baseline featured in section 4.1, with a similar secular decline in the central density. This is a key result because it suggests that the mechanism behind the decline does not require the full long-range potential structure of the halo, and that truncating interactions beyond 10 kpc does not eliminate the effect.

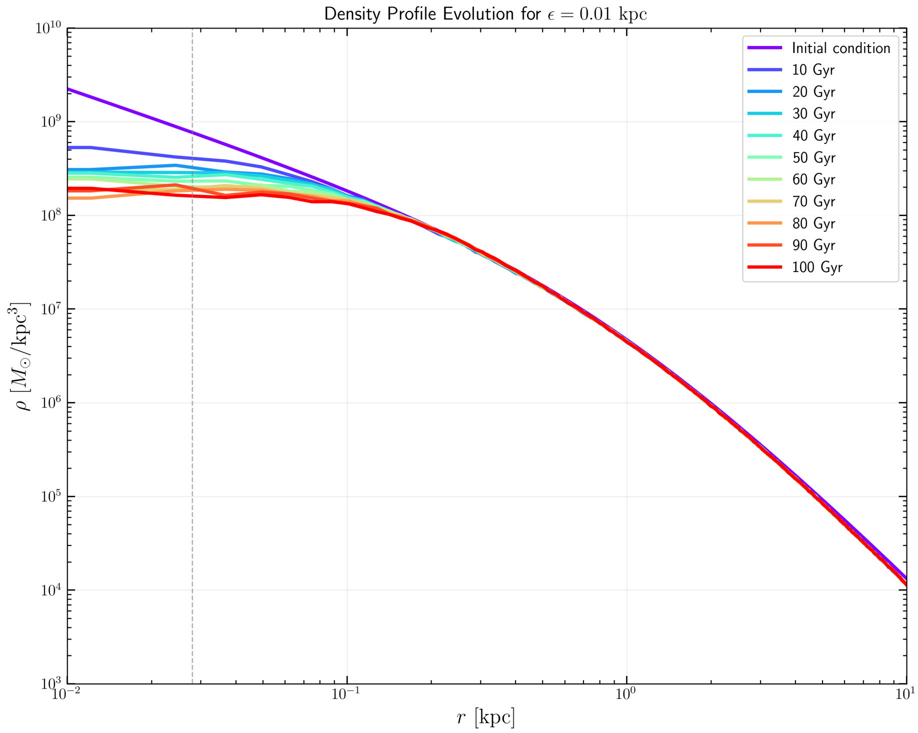


Figure 4.12: **Baseline density-profile evolution.** The central density decreases steadily over time, with the steepest change occurring early and the later evolution approaching a shallower, but still declining, trend. The gradient structure is consistent with a secular redistribution rather than a settled core.

This motivates further analysis of smaller Rcut values, which is carried out in the following plot.

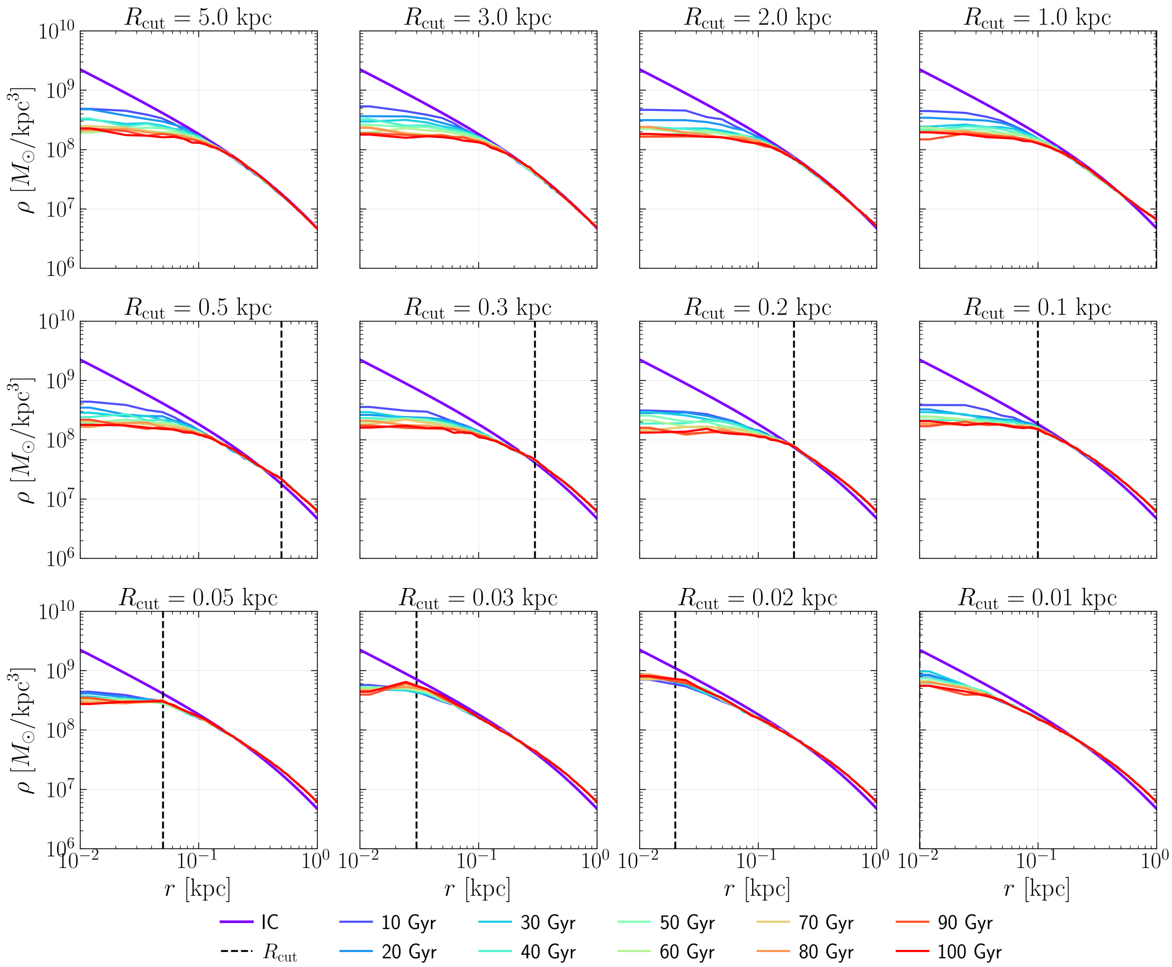


Figure 4.13: **Rcut study large plot.**

The figure above provides an overview of simulations with small $R_{\rm cut}$ values. For $R_{\rm cut} \gtrsim 0.1$ kpc, the system evolves very similarly to the reference (vanilla SWIFT) run. However, as $R_{\rm cut}$ approaches the softening length value, the cusp-to-core transition does not extend to large times.

This behaviour is clearly illustrated in the following figures: while the simulation with $R_{\rm cut} = 5$ kpc exhibits continued core growth up to 100 Gyr, progressively smaller values of $R_{\rm cut}$ lead to increasingly limited core evolution, eventually resulting in a density profile that does not decrease.

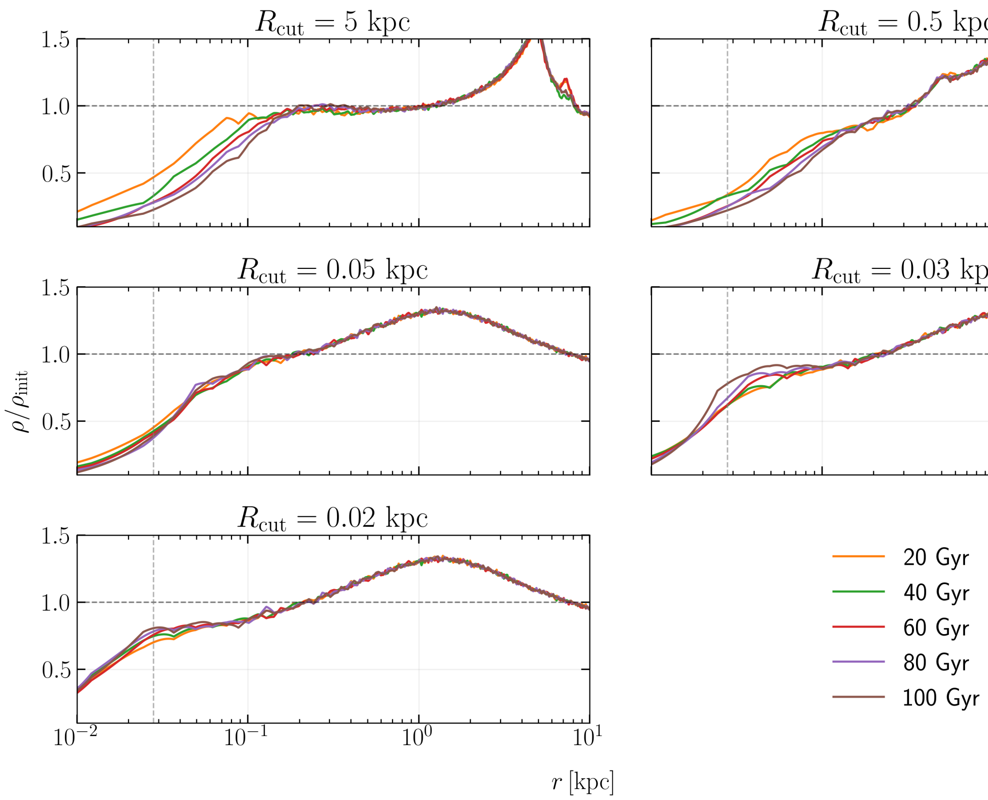


Figure 4.14: **Density ratio comparison**

Complete plots can be found in the annex Section D.

Finally, the results are summarized in Figure 4.15, which compares the density evolution for multiple R_{cut} values at 50 Gyr and 100 Gyr. The left panel shows that at 50 Gyr, the runs with $R_{\text{cut}} \gtrsim 0.1$ kpc are nearly indistinguishable from the baseline, while smaller cutoffs show a more limited decline. The right panel confirms that by 100 Gyr, the same hierarchy persists.

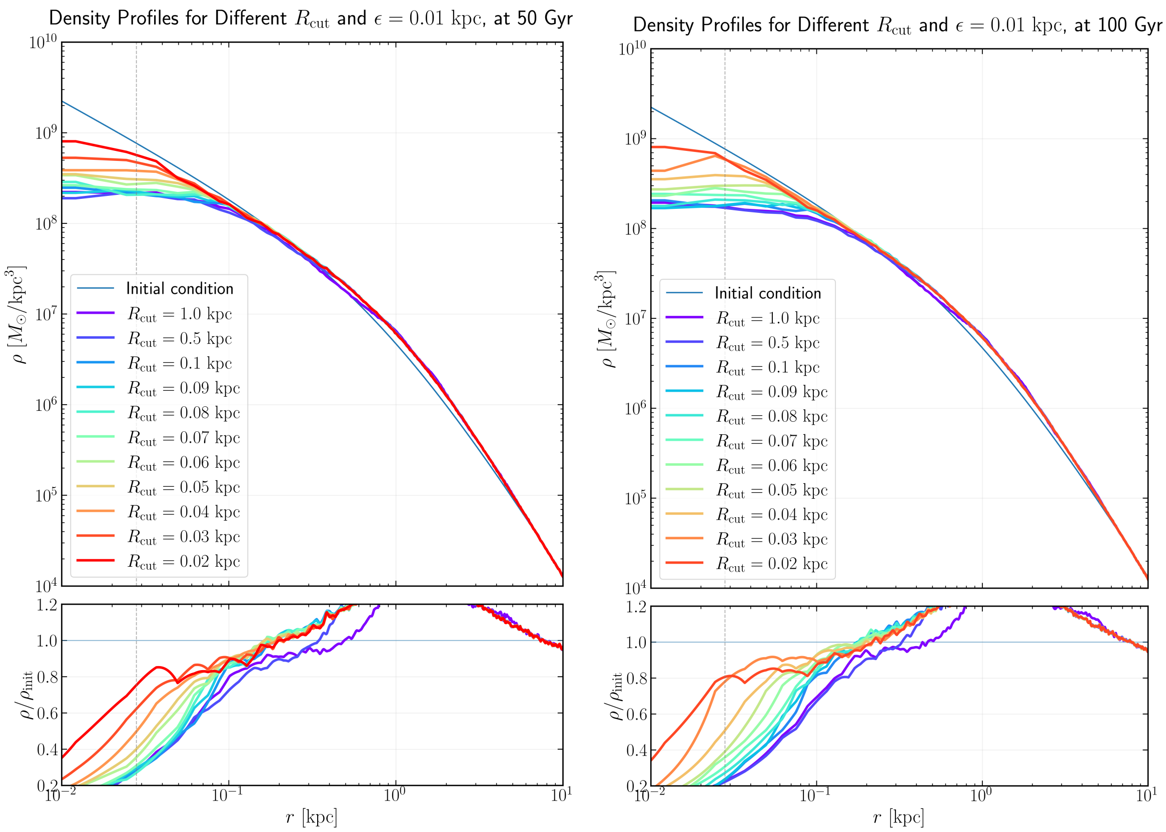


Figure 4.15: **Rcut comparison.**

These results indicate that the central density decline is driven by a mechanism that does not require the full long-range potential structure of the halo, and that truncating interactions beyond 10 kpc does not eliminate the effect. However, as the cutoff radius approaches the softening length, the core evolution becomes increasingly limited, suggesting that the mechanism is sensitive to interactions at scales comparable to the softening length.

The result indicates that the late evolution requires a sufficiently extended self-gravitating environment, but it does not show that the outer halo is directly heating the center. It may simply be acting as the reservoir that sustains the discreteness noise or orbital diffusion.

Artificial two-body relaxation and force softening appear to be driving the persistent core formation in the halo simulations. Studies of high-resolution CDM halos show that even 10^6–10^8 particles have relaxation times short enough to flatten cusps in a few dynamical times. In particular, Diemand et al. (2004) demonstrated that simulated halos suffer central heating that declines only as $\sim N^{-0.25}$ with particle number, rather than N^{-1}, implying that large N ($\gg 10^8$) or substantial softening are needed to prevent cusp erosion. Similarly, El-Zant (2006) finds that $t_{\rm relax} \propto r^2$ and that a 10^6-particle halo is already "noise-dominated" in the inner $\sim 1\%$, requiring $N \gg 10^8$ for the inner profile to remain collisionless. These works predict that truncating outer mass alone cannot halt inner relaxation:

removing long-range forces beyond ~ 10–20 kpc still leaves the small-N central cusp relaxationally unstable. In contrast, the softening length must be large enough to suppress close-encounter scattering. In short, the balance of evidence is that numerical discreteness (two-body scattering and shot noise), rather than algorithmic bias or mis-centering, is the primary culprit.

5 Conclusions

Bibliography

AAS Nova (2024). Down to the core: Dark matter deviations in ultra-faint dwarf galaxies. https://aasnova.org/2024/11/01/down-to-the-core-dark-matter-deviations-in-ultra-faint-dwarf-galaxies/#prettyPhoto. Accessed: 2026-05-21.

Binney, J. and Tremaine, S. (2008). *Galactic Dynamics*. Princeton University Press, Princeton, NJ, 2nd edition.

Boylan-Kolchin, M., Springel, V., White, S. D. M., Jenkins, A., and Lemson, G. (2009). Resolving cosmic structure formation with the millennium-ii simulation. *Monthly Notices of the Royal Astronomical Society*, 398:1150–1163. arXiv:0903.3041 [astro-ph].

Crnojević, D. and Mutlu-Pakdil, B. D. (2021). Dwarf galaxies yesterday, now and tomorrow. *Nature Astronomy*, 5:1191–1194.

Diemand, J., Moore, B., Stadel, J., and Kazantzidis, S. (2004). Two body relaxation in CDM simulations. *Monthly Notices of the Royal Astronomical Society*, 352:535–546. arXiv:astro-ph/0304549.

El-Zant, A. A. (2006). Two body relaxation in simulated cosmological haloes. *Monthly Notices of the Royal Astronomical Society*, 370:1475–1485. arXiv:astro-ph/0506617.

Errani, R., Esser, N., Peñarrubia, J., and Walker, M. G. (2026). Collisional dynamics of stars and dark matter in ultra-faint galaxies. *The Astrophysical Journal, submitted*. arXiv:2604.06304.

Hopkins, P. F., Nadler, E. O., Grudić, M. Y., Shen, X., Sands, I., and Jiang, F. (2023). Novel conservative methods for adaptive force softening in collisionless and multi-species N-body simulations. *Monthly Notices of the Royal Astronomical Society*. arXiv:2212.06851v2.

Navarro, J. F., Frenk, C. S., and White, S. D. M. (1996). The structure of cold dark matter halos. *The Astrophysical Journal*, 462. arXiv:astro-ph/9508025.

Peñarrubia, J. (2017). Fluctuations of the gravitational field generated by a random population of extended substructures. *Monthly Notices of the Royal Astronomical Society*. arXiv:1710.06443v2.

Peñarrubia, J. (2019). Stochastic tidal heating by random interactions with extended substructures. *Monthly Notices of the Royal Astronomical Society*. arXiv:1901.11536v1.

Power, C., Navarro, J. F., Jenkins, A., Frenk, C. S., White, S. D. M., Springel, V., Stadel, J., and Quinn, T. (2003). The inner structure of λCDM haloes – I. A numerical convergence study. *Monthly Notices of the Royal Astronomical Society*, 338(1):14–34.

Price, D. J. and Monaghan, J. J. (2007). An energy-conserving formalism for adaptive gravitational force softening in SPH and N-body codes. *Monthly Notices of the Royal Astronomical Society*, pages 1–13. arXiv:astro-ph/0610872.

Salamin, G. G. H. (2026). The origin of size growth in ultra-faint dwarf galaxies: a comparison of cold and warm dark matter zoom-in N-body simulations. Master's thesis, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland.

SWIFT Collaboration (2024). Swift simulation gallery. https://swift.strw.leidenuniv.nl/gallery.html. Accessed: 2026-05-21.

Zhang, T., Liao, S., Li, M., and Gao, L. (2019). The optimal gravitational softening length for cosmological N-body simulations. *Monthly Notices of the Royal Astronomical Society*, 487(1):1227–1243.

A Shrinking sphere centering method

This appendix gives the mathematical formulation and a direct algorithmic implementation of the shrinking-sphere centering routine used in the analysis.

Let the particle set be indexed by i, with positions $\mathbf{x}_i \in \mathbb{R}^3$ and masses $m_i > 0$. Define the mass-weighted centroid over an index set S as

$$\mathbf{C}(S) = \frac{\sum_{i \in S} m_i \mathbf{x}_i}{\sum_{i \in S} m_i}. \tag{A.1}$$

Initialize

$$S_0 = \{1, \ldots, N\}, \qquad \mathbf{x}_0 = \mathbf{C}(S_0), \qquad R_0 = \max_{i \in S_0} \|\mathbf{x}_i - \mathbf{x}_0\|. \tag{A.2}$$

For iterations $k = 0, 1, 2, \ldots$ define the restricted set and update rules

$$S_{k+1} = \{i \in S_k : \|\mathbf{x}_i - \mathbf{x}_k\| < R_k\}, \tag{A.3}$$

$$\mathbf{x}_{k+1} = \mathbf{C}(S_{k+1}), \tag{A.4}$$

$$R_{k+1} = f_s R_k, \tag{A.5}$$

with shrink factor $0 < f_s < 1$ (implementation default $f_s = 0.9$).

Stop when either

$$\|\mathbf{x}_{k+1} - \mathbf{x}_k\| < \delta \quad \text{or} \quad |S_{k+1}| < N_{\min}, \tag{A.6}$$

and return $\mathbf{x}_{\text{final}} = \mathbf{x}_{k+1}$. Typical choices used in the analysis are $\delta = 10^{-5}$ (code length units) and $N_{\min} = 100$.

The same logic can be written in compact pseudocode as follows:

```
function shrinking_sphere_center(pos[N,3], mass[N], f_s=0.9, delta=1e-5, N_min=100):
    x = weighted_average(pos, mass)
    R = max(norm(pos - x, axis=1))
```

```
    while True:
        r = norm(pos - x, axis=1)
        mask = (r < R)
        if mask.sum() < N_min:
            break
        x_new = weighted_average(pos[mask], mass[mask])
        if norm(x_new - x) < delta:
            x = x_new
            break
        x = x_new
        R = R * f_s
    return x
```

B Analytic shell decomposition

The analysis uses a spherically symmetric Navarro–Frenk–White (NFW) halo model (Navarro et al., 1996), which is a standard benchmark for dark matter halos:

$$\rho(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \tag{B.1}$$

where ρ_s is the characteristic density and r_s is the scale radius. The enclosed mass within radius r is:

$$M(< r) = 4\pi\rho_s r_s^3 \left[\ln(1 + x) - \frac{x}{1 + x}\right], \quad x = \frac{r}{r_s}. \tag{B.2}$$

The halo is truncated at a virial radius r_{vir}. The gravitational potential including the truncation is:

$$\Phi(r) = -\frac{GM(< r)}{r} - 4\pi G\rho_s r_s^2 \left[\frac{1}{1 + r/r_s} - \frac{1}{1 + r_{\mathrm{vir}}/r_s}\right], \tag{B.3}$$

and the radial acceleration (force per unit mass) is:

$$a(r) = \frac{GM(< r)}{r^2}. \tag{B.4}$$

B.1 Key diagnostics and definitions

The analysis is built around four central diagnostic quantities, each addressing a different aspect of halo structure:

Enclosed mass and shell mass fraction The enclosed mass $M(< r)$ tracks how the total mass budget grows with radius. The shell mass fraction is the mass in a radial shell divided by the total halo mass, showing the radial distribution of the mass budget. For an NFW profile, the enclosed mass grows steeply near the center but the shell mass fraction increases toward larger radii, indicating that the outer halo contains a substantial fraction of the total mass.

Central potential depth from individual shells For each shell at radius range $[r_i, r_{i+1}]$, we calculate its contribution to the gravitational potential at the center ($r = 0$). This is given exactly by:

$$\Delta\Phi_{\rm shell}(0) = -4\pi G\rho_s r_s^2 \left[\frac{1}{1+r_i/r_s} - \frac{1}{1+r_{i+1}/r_s}\right]. \tag{B.5}$$

Shells well outside the center still make sizeable contributions to the potential depth at $r = 0$ (and therefore to the escape-speed scale $v_{\rm esc} \propto \sqrt{-2\Phi}$). The fractional contribution of each shell quantifies this effect.

Outer-potential fraction For a given probe radius $r_{\rm probe}$ and cutoff radius $r_{\rm cut}$, the outer-potential fraction is defined as:

$$f_{\rm outer}(r_{\rm probe}, r_{\rm cut}) = \frac{\left|\Phi_{>r_{\rm cut}}(r_{\rm probe})\right|}{\left|\Phi_{\rm total}(r_{\rm probe})\right|}, \tag{B.6}$$

where $\Phi_{>r_{\rm cut}}$ is the potential contribution from material outside the cutoff and $\Phi_{\rm total}$ is the total potential at that radius. This measures how strongly the outer halo influences the local potential at the probe radius. By varying $r_{\rm cut}$, we can determine the radius beyond which the outer halo becomes a small correction.

Force contribution by shell The radial force (acceleration) at a probe radius $r_{\rm probe}$ is:

$$a(r_{\rm probe}) = \frac{GM(< r_{\rm probe})}{r_{\rm probe}^2}. \tag{B.7}$$

Under spherical symmetry, only mass interior to $r_{\rm probe}$ contributes. This diagnostic confirms that outer shells, despite deepening the potential, do not directly modify the local force field—a consistency check on the shell-theorem interpretation.

B.2 Implementation: analytical NFW shell decomposition

Rather than relying on a numerical computation of the integrals, the shell analysis uses exact analytic formulas for a truncated NFW profile. The halo is decomposed into logarithmically spaced spherical shells:

$$r_i = r_{\rm min}\left(\frac{r_{\rm vir}}{r_{\rm min}}\right)^{i/N_{\rm shells}}, \quad i = 0, 1, \ldots, N_{\rm shells}, \tag{B.8}$$

where typically $N_{\text{shells}} = 80$ and $r_{\min} \approx 0.02$ kpc.

For each shell $[r_i, r_{i+1}]$, exact closed-form expressions compute: (1) shell mass $M_{\text{shell}} = M(< r_{i+1}) - M(< r_i)$, (2) shell contribution to potential at any radius, and (3) cumulative potential contributions for integration over shells.

The analytical approach avoids numerical-integration noise and ensures all diagnostics are computed to high precision. The Python implementation uses these formulas to generate diagnostic plots showing: (i) enclosed mass and shell mass-fraction distribution, (ii) shell contributions to the central potential depth, (iii) outer-potential fraction as a function of cutoff radius, and (iv) a 2D map of outer-fraction versus both probe and cutoff radii.

B.3 Results

B.3.1 Mass and potential distribution across the halo

The first two diagnostics establish the overall radial structure of the halo. Figure B.1 shows the enclosed mass profile and the fractional mass distribution per shell.

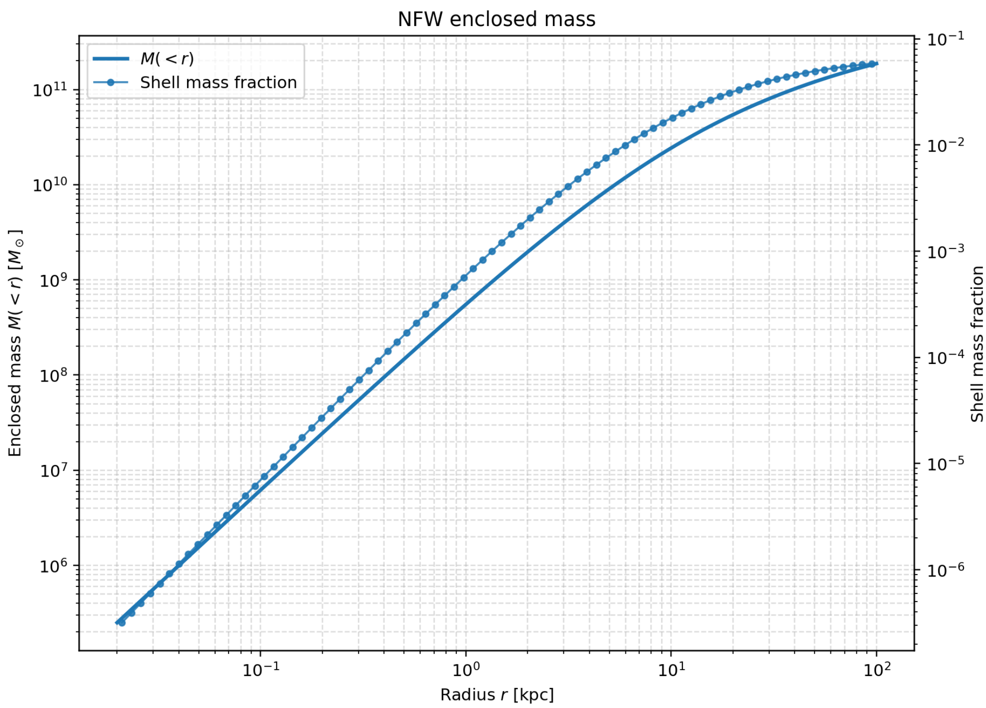


Figure B.1: **NFW enclosed mass and shell mass fraction.** Left axis: monotonically increasing enclosed mass $M(< r)$ with radius. Right axis: shell mass fraction per logarithmic shell.

The enclosed-mass profile follows the expected NFW form: steep near the center and flattening at larger radii. Importantly, the shell mass fraction plot shows that although the inner shells contain many particles per unit volume (high density), the integrated mass grows progressively outward. Shells at radii $r > 20$ kpc contain a cumulatively significant mass fraction, establishing that the outer halo is not negligible.

Figure B.2 shows the contribution of each shell to the gravitational potential at the center and the cumulative contribution from all shells outside a given radius.

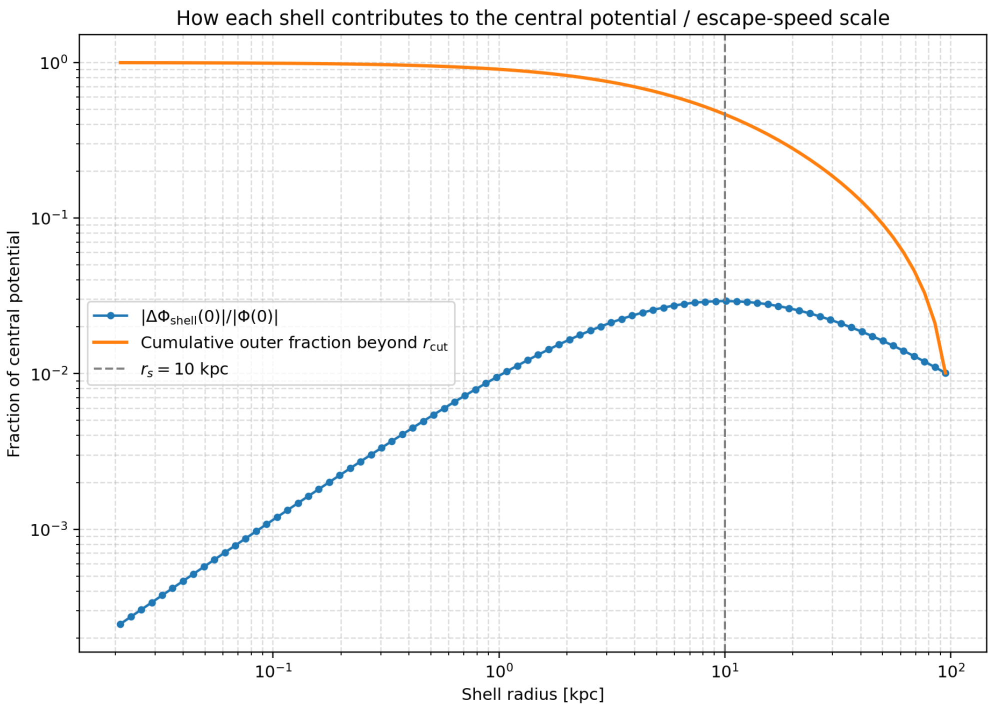


Figure B.2: **Shell contributions to the central potential depth.** Individual shells (blue points with error bars) contribute to the total potential depth $|\Phi(0)|$. The cumulative contribution (orange line) from all shells beyond a given radius shows that shells at radii beyond the scale radius r_s still make substantial contributions to the escape-speed scale.

The key finding here is that shells well outside the center, at radii several times the scale radius, still contribute visibly to the potential depth. The fractional contribution decreases with radius but remains non-negligible up to $r \sim 3r_s$. This result motivates the choice of cutoff radius in the hybrid model; setting r_{cut} too small would omit a significant part of the potential structure.

B.3.2 Outer-halo influence as a function of cutoff radius

The most critical diagnostic for the hybrid model design is the outer potential fraction. The fraction of the total potential at a given probe radius that arises from material beyond a chosen cutoff radius. Figure B.3 shows this quantity for several probe radii as the cutoff radius is increased.

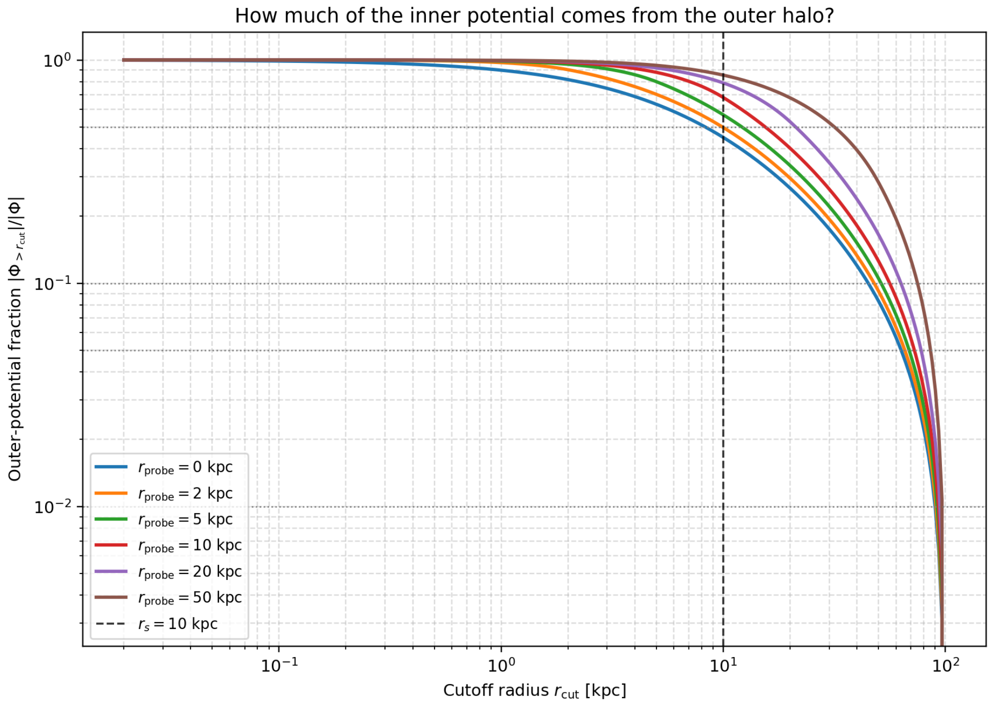


Figure B.3: **Outer-potential fraction versus cutoff radius.** For probe radii ranging from the center to $r = 5r_s$, the fraction of the potential originating from material beyond the cutoff radius r_{cut} is plotted against the cutoff. As the cutoff moves outward, the outer contribution drops sharply initially, then plateaus. For the central region ($r_{probe} < r_s$), a cutoff at $r_{cut} \approx 10$ kpc retains roughly 5–10% of the potential from the outer halo; at $r_{cut} \approx 30$ kpc, this drops below 1%. Thus, if the outer halo contributes significantly to the potential at the radii of interest (e.g., $r < 1$ kpc for the central density), then the outer region must be represented accurately.

The curves in this figure answer the central question: how large must r_{cut} be to capture the essential potential structure? For the central region, a cutoff at $r_{cut} = 10$ kpc retains the majority of the potential at small radii while excluding the outermost, kinematically less important material. This is a quantitative justification for the parameter choice in the numerical hybrid experiment.

B.3.3 2D outer-fraction map

Figure B.4 provides a compact summary of the outer-potential fraction across all combinations of probe and cutoff radii.

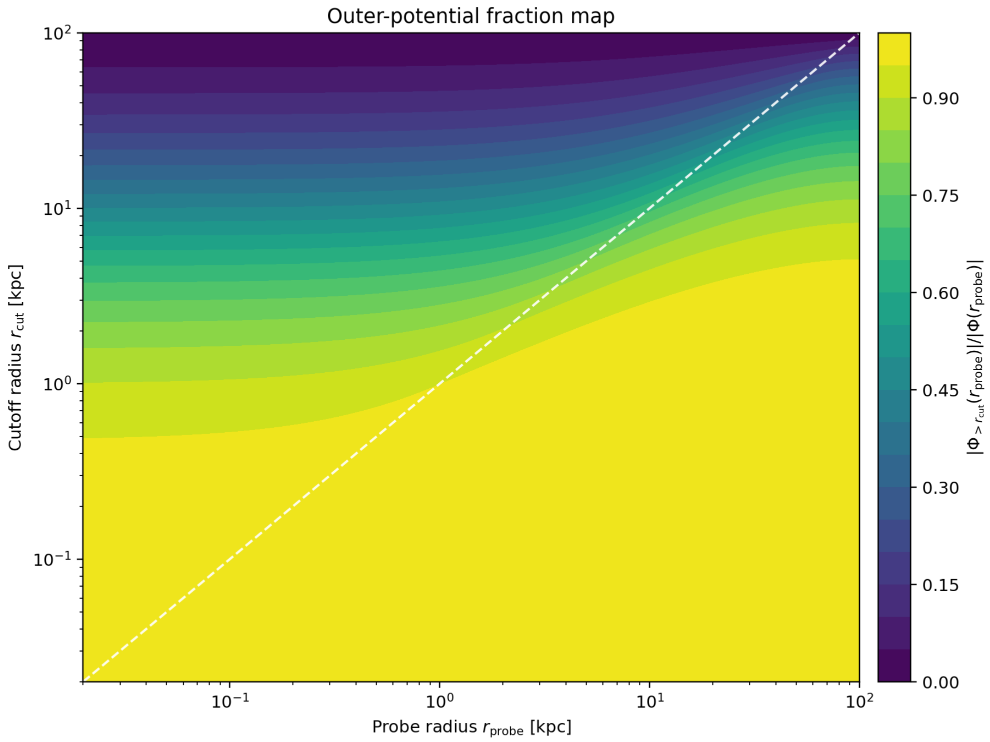


Figure B.4: **2D map of the outer-potential fraction.** Rows represent probe radii; columns represent cutoff radii. The color intensity indicates the fraction of potential at a given probe radius arising from material beyond the cutoff. The transition from high (red) to low (blue) values shows the radius beyond which the outer halo becomes a small correction.

The 2D map reveals the structure clearly: for small cutoff radii (lower rows), a broad range of inner probe radii feel substantial outer halo influence. As the cutoff moves outward, this influence shrinks to a narrow band near the cutoff itself.

C Orbits

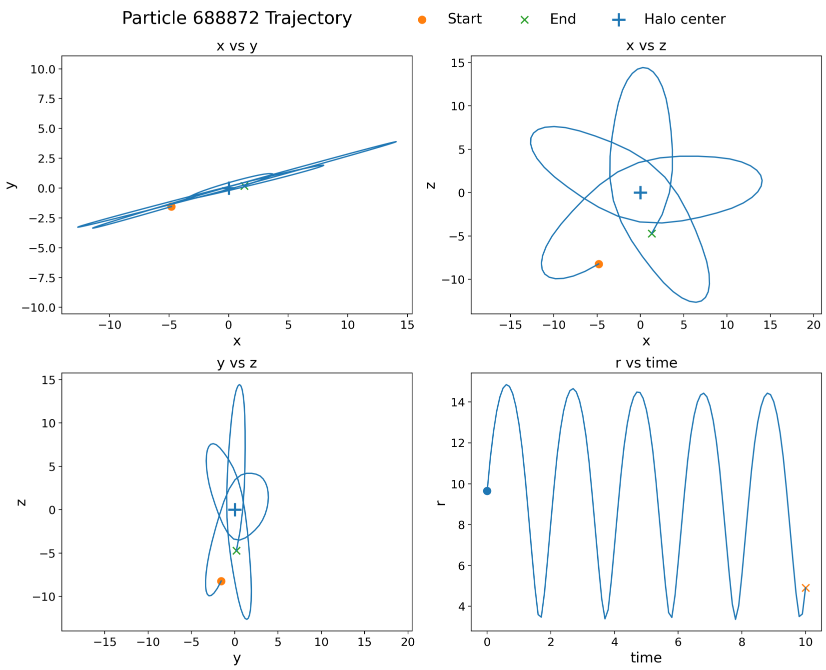


Figure C.1: Orbit of selected particle 1

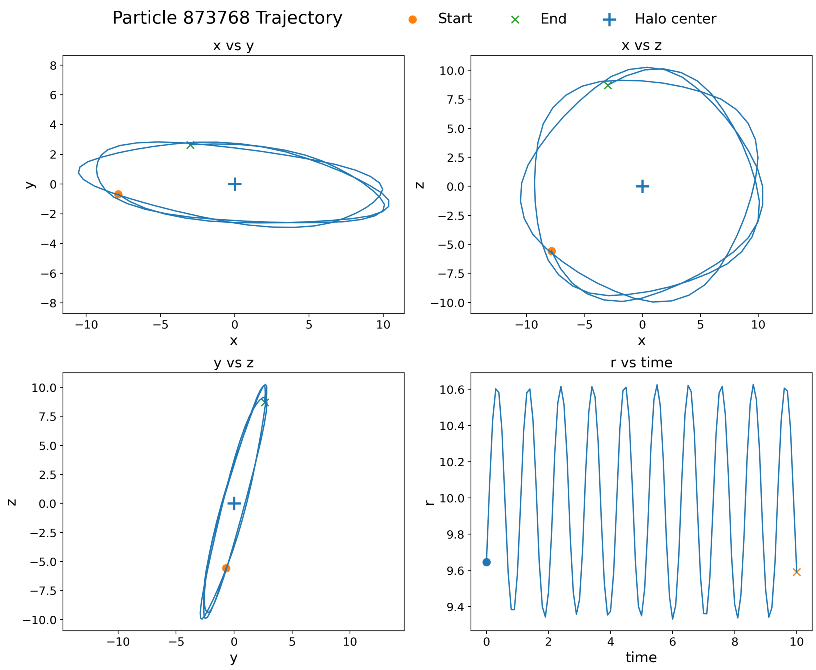


Figure C.2: Orbit of selected particle 2

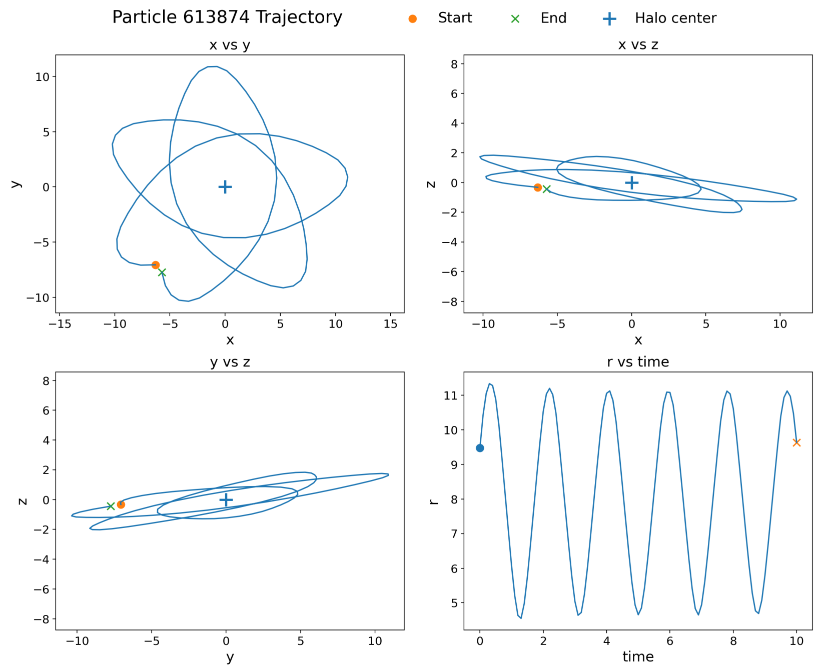


Figure C.3: Orbit of selected particle 3

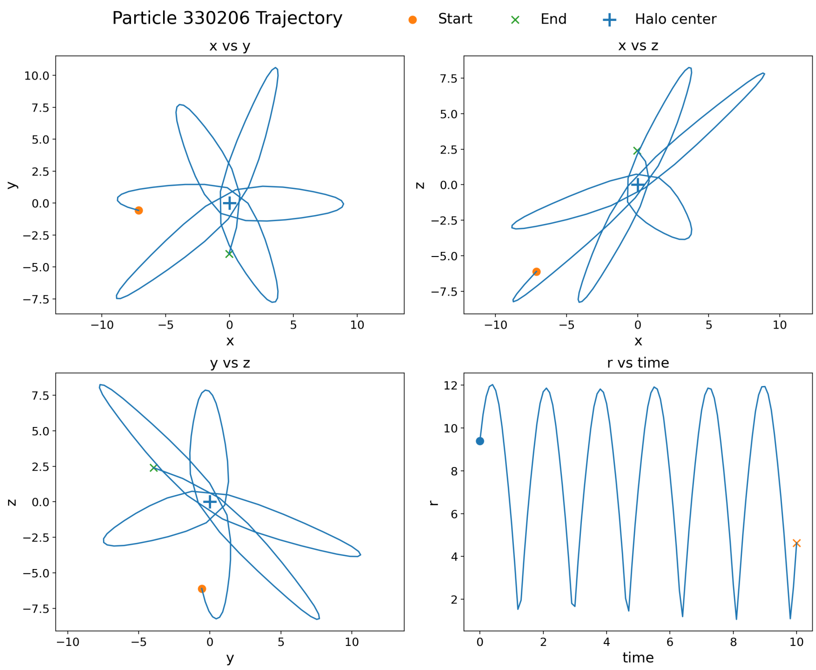


Figure C.4: Orbit of selected particle 4

D Truncated-NFW density profile evolution

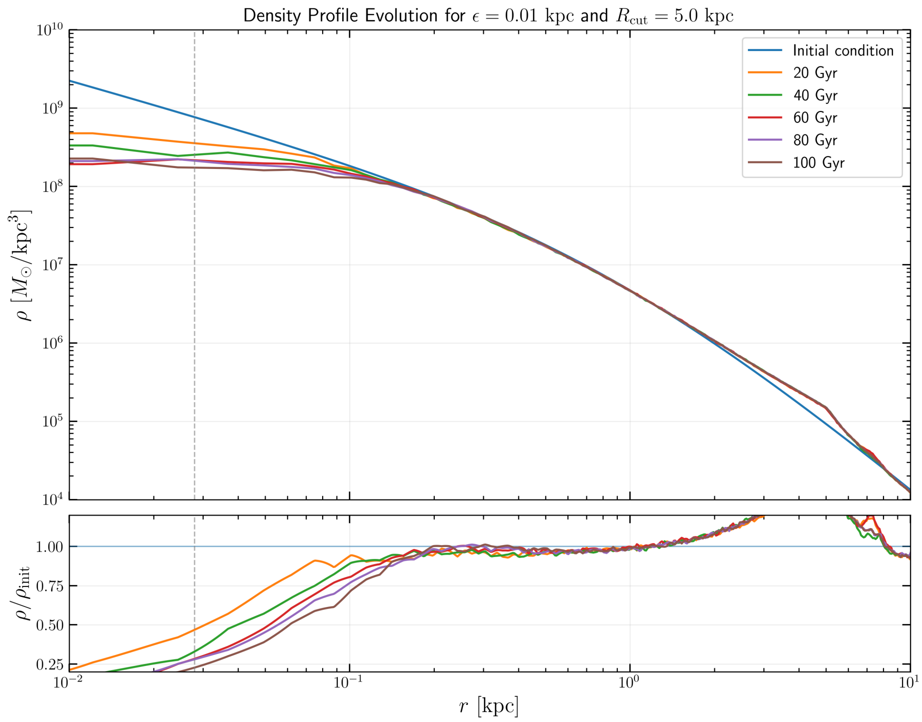


Figure D.1: Density profile evolution for the truncated-NFW experiment with cutoff at 5 kpc.

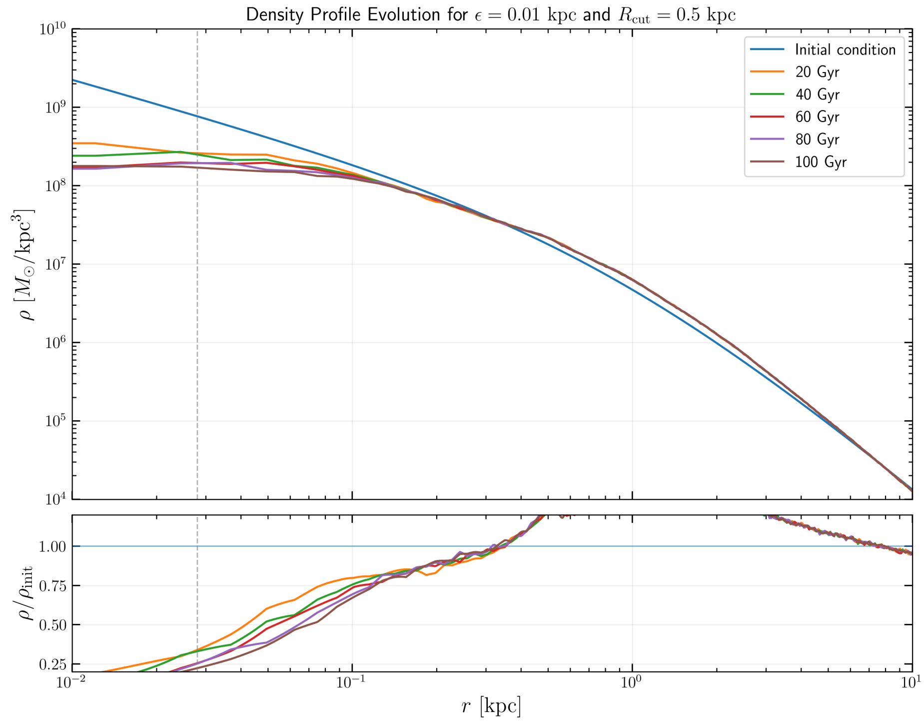


Figure D.2: Density profile evolution for the truncated-NFW experiment with cutoff at 0.5 kpc.

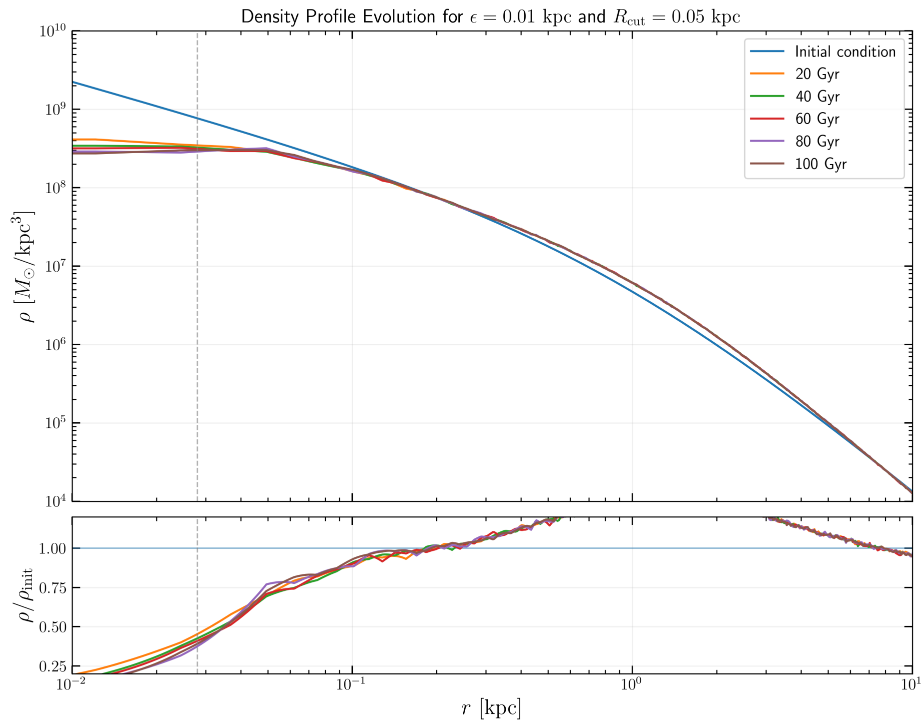


Figure D.3: Density profile evolution for the truncated-NFW experiment with cutoff at 0.05 kpc.

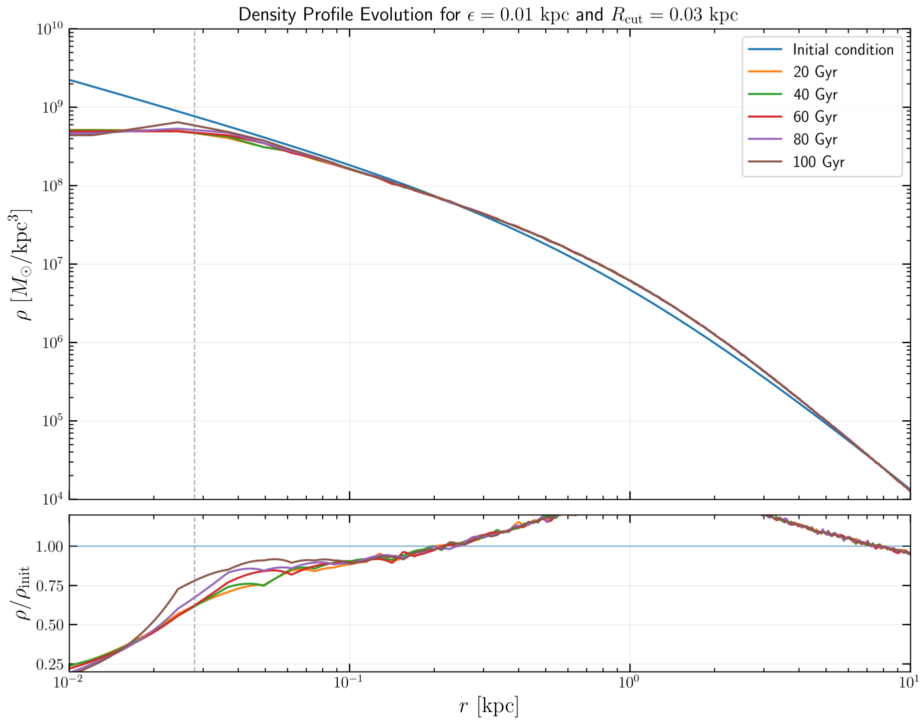


Figure D.4: Density profile evolution for the truncated-NFW experiment with cutoff at 0.03 kpc.

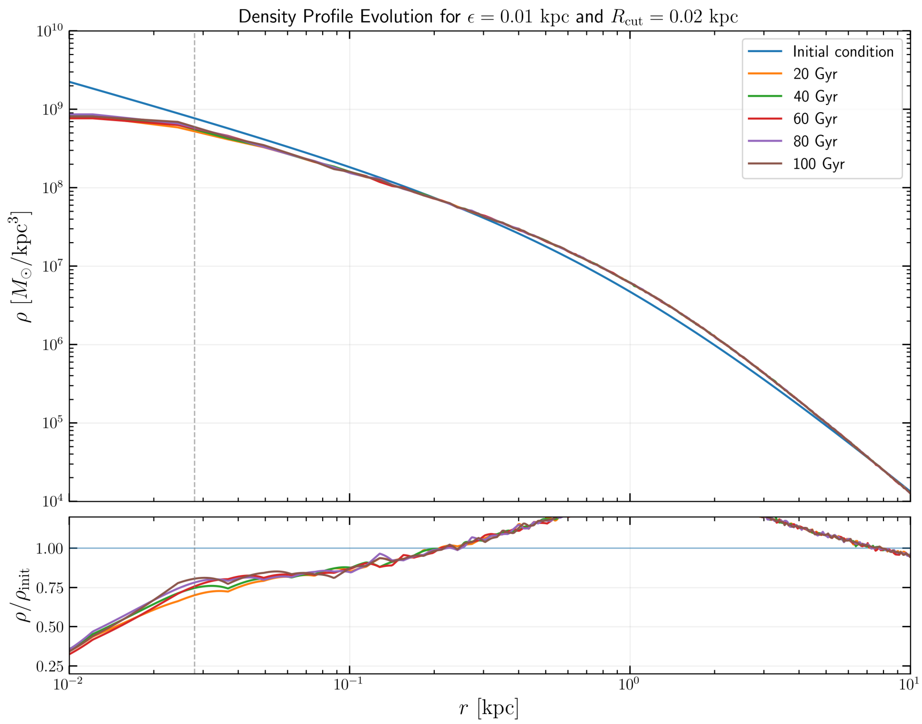


Figure D.5: Density profile evolution for the truncated-NFW experiment with cutoff at 0.02 kpc.